

open-media-drc

Digital Room Correction media chain for Linux and FreeBSD — User and Administrator Manual

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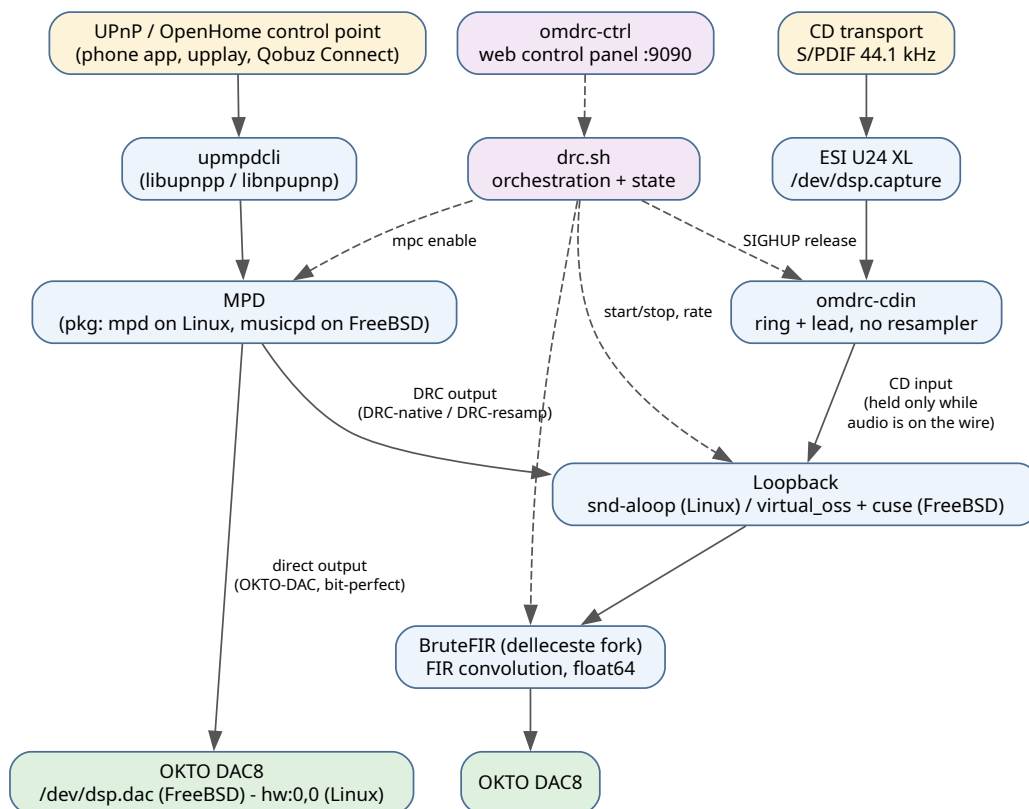
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1 Introduction

open-media-drc is the complete software stack of a headless, high-quality music and video playback appliance with **Digital Room Correction (DRC)**. It runs on both **Linux** (reference: Arch) and **FreeBSD** (reference: 15.1 on an Intel NUC), driving an OKTO RESEARCH DAC8 STEREO USB DAC.

The core idea: audio from any source — UPnP/OpenHome streaming (Qobuz), local files, Blu-ray discs, network streams — is routed through **BruteFIR**, a fast FIR convolution engine, which applies room-correction filters designed with Room EQ Wizard (REW) and DRC tooling, before reaching the DAC. When correction is off, the chain collapses to a verified **bit-perfect** direct path.



The full audio playback chain, from control point to DAC. Solid arrows carry audio; dashed arrows are control.

The design principles that shape everything in the repository:

- **Saved user intent plus actual device state drive DRC.** A present DAC and saved power-on intent mean DRC up at the saved source/rate/design; an absent DAC means transient teardown without losing that intent; saved power-off means direct output. Boot, hotplug, and manual requests reconcile this rule.
- **Only two resting states exist:** DRC fully up (BruteFIR processing), or direct output. A failed start rolls back — there is no resting state where the loopback runs without BruteFIR.
- **Run-from-repo.** The live configuration files and scripts are symlinks into the git checkout wherever possible; `git pull` is the whole update path. Only “deploy glue” parsed at early boot (systemd units, udev rules, rc.d scripts, devd rules) is *copied* into system paths, because those are parsed before a separately mounted `/home` is available.
- **Verify, don’t assume.** The repository ships tooling to *prove* the chain is bit-perfect (a USB wire tap), to detect and classify glitches, and to monitor the whole chain from a phone.

1.1 Repository map

Path	Contents
drc.sh, drc-status.sh	DRC orchestration and status
configs/<geometry>/	Per-rate BruteFIR configurations
filters/<geometry>/<rate>/	Raw FIR filter coefficients (FLOAT64_LE)
mpd/	MPD configuration templates (mpd.conf.in Linux, musicpd.conf.in FreeBSD)
etc/	Service glue: systemd/, modules-load.d/ (Linux); rc.d/, devd/ (FreeBSD)
omdrc-ctrl/	Web control panel (Flask)
cdin/	omdrc-cdin, the CD / S-PDIF capture bridge (FreeBSD only)
video/	mpv playback launchers + phone web remote
browser-nodrc/	Browser launchers that temporarily bypass DRC
scripts/	Filter conversion, headroom, verification helpers
tests/	Bit-perfect test signal
freebsd-uaudio-patch/	FreeBSD kernel uaudio(4) patches (Appendix A)
freebsd-virtual-oss-patch/	FreeBSD virtual_oss / cuse patches (Appendix A)
kodi-virtual-oss-patch/	Kodi OSS-sink enumeration patch (Appendix A)
doc/	Measurement plots, verification and glitch docs, this manual

2 Components

The playback chain is assembled from the following components, in signal order.

2.1 upmpdcli, libupnp, libnpupnp (UPnP/OpenHome front-end)

`upmpdcli` turns MPD into a UPnP/OpenHome media renderer, so any control point (a phone app, `uplay`, Qobuz Connect) can drive playback. It is built from source, bottom-up, as three standard meson projects: **libnpupnp** (UPnP base library; needs `libcurl`, `libmicrohttpd`, `expat`), **libupnpp** (C++ wrapper), then **upmpdcli** itself (needs `jsoncpp`, `libmpdclient`). The optional Qobuz plugin needs `python3` with `requests`. Prebuilt packages exist (FreeBSD port `upmpdcli`, Arch AUR); source builds are used here to track upstream.

2.2 MPD — Music Player Daemon

The actual player. Installed **from the OS package**:

Naming: MPD is packaged as `mpd` on Linux, but as `musicpd` on FreeBSD. The FreeBSD package/port is `audio/musicpd`, the daemon binary is `musicpd`, the service is `service musicpd` . . . , and the bundled command-line client is `musicpc` (aliasing `mpc`). Everywhere this manual says “MPD”, read `mpd` on Linux and `musicpd` on FreeBSD.

MPD must have the **soxr** resampler and the ALSA (Linux) / OSS (FreeBSD) output plugins enabled — both stock packages do. MPD is configured with three outputs (section): the direct DAC output and two DRC loopback outputs.

2.3 omdrc-cdin — the CD / S-PDIF capture bridge (FreeBSD)

The second source, and the only live one: a CD transport’s S/PDIF output, captured through an ESI U24 XL and written into the same loopback MPD uses. It is a `memcpy` through a ring whose *lead* absorbs the permanent few-ppm difference between the disc’s crystal and the DAC’s — no resampler, so the bit-perfect claim survives the CD path too. It holds the loopback only while audio is actually on the wire, so leaving it running costs MPD and `mpv` nothing. Full treatment, including the ESI configuration it depends on, in chapter .

2.4 The loopback: `snd-aloop` (Linux) / `virtual_oss` (FreeBSD)

BruteFIR needs to receive the player’s audio. That is done with a loopback device MPD plays into and BruteFIR reads from:

- **Linux:** the `snd-aloop` ALSA kernel module (loaded via `etc/modules-load.d/`).
- **FreeBSD:** **`virtual_oss`**, a userland OSS mixing/routing daemon from the base system, which creates character devices through the **`cuse(3)`** kernel facility. `drc.sh` starts it per rate with a play node (`/dev/dsp.play`, where MPD writes) and a synchronized loopback node (`/dev/dsp.loop`, where BruteFIR reads): the `-L` loopback. The `cuse` kernel module must be loaded.

2.5 BruteFIR (delleceste fork)

The convolution engine, built from github.com/delleceste/brutefir — a fork of Anders Torger's classic BruteFIR adding FreeBSD OSS fixes (bfio_oss fragment-size fix, brutefir_loopback -L fix, and a passthrough-config default). It processes audio entirely in **float64**, convolving the per-rate L . raw/R . raw FIR filters and writing to the DAC. Requires FFTW3 in both single and double precision; ALSA on Linux (OSS support is built in on FreeBSD). Built with CMake; modules install to `/usr/local/lib/brutefir`.

2.6 open-media-drc proper: drc.sh and the configuration tree

`drc.sh` is the single control point of the DRC pipeline: it starts and stops BruteFIR and the loopback, selects the MPD output, primes the DAC on rate changes, serializes concurrent runs, records persistent state, and rolls back to direct output on failure. Section covers it in full.

2.7 omdrc-ctrl — web control panel

A lightweight Flask web app serving a mobile-friendly, dark, touch-first control panel on the LAN (default port 9090). It exposes DRC control buttons, full audio-chain health monitoring with a bit-perfect verdict, a live spectrum analyzer, and DRC filter-response charts. Section .

2.8 video/ — mpv playback and the phone web remote

Blu-ray discs, DVDs, files and streams played through mpv with audio routed through the same DRC chain, controlled either from KDE Connect (MPRIS) or from a dedicated phone web remote (the mpv-only Kodi replacement). Section .

2.9 browser-nodrc — DRC bypass for browsers

Web browsers cannot easily route into the DRC loopback, and BruteFIR holds the DAC single-open while DRC runs. `browser-nodrc/` provides one launcher per browser (Firefox, Chromium, Chrome) that snapshots the DRC state, runs `drc.sh off`, launches the browser in the foreground, and restores the exact pre-launch state from an EXIT trap — even if the browser crashes.

3 Installation

3.1 Dependencies

Build tools for all from-source components: a C/C++ compiler, **meson + ninja** (upmpdcli stack), **cmake** (BruteFIR fork, omdrc-ctrl), **pkg-config**, and git.

Component	Runtime deps	FreeBSD pkg	Arch pacman
libnpupnp 6.3.0	libcurl, libmicrohttpd, expat	curl libmicrohttpd expat2	curl libmicrohttpd expat
libupnpp 1.0.4	libnpupnp + the above	(above)	(above)
upmpdcli 1.9.17	libupnpp, jsoncpp, libmpdclient	jsoncpp libmpdclient	jsoncpp libmpdclient
upmpdcli Qobuz plugin	python3 + requests	python3 py311-requests	python python-requests
MPD	soxr resampler + ALSA/OSS output	musicpd	mpd
BruteFIR (fork)	FFTW3 single+double; ALSA on Linux	fftw3 fftw3-float	fftw alsa-lib
Loopback	—	virtual_oss (+ cuse)	snd-aloop kernel module
omdrc-ctrl	python3, flask>=2.3, markdown>=3.5, numpy>=1.21 (optional)	py311-flask py311-Markdown py311-numpy	python-flask python-markdown python-numpy

3.2 Build and install order

1. The upmpdcli stack (bottom-up; each a standard meson project):

```
for p in libnpupnp-6.3.0 libupnpp-1.0.4 upmpdcli-1.9.17; do
  cd ~/Downloads/$p
  meson setup build --prefix=/usr/local
  ninja -C build
  sudo ninja -C build install
done
sudo ldconfig 2>/dev/null || true # Linux: refresh the linker cache
```

2. MPD — from the OS package:

```
sudo pkg install musicpd # FreeBSD
sudo pacman -S mpd # Arch Linux
```

3. BruteFIR — from the fork:

```
git clone https://github.com/delleceste/brutefir ~/Downloads/brutefir
cd ~/Downloads/brutefir
cmake -B build && cmake --build build
sudo cmake --install build # modules -> /usr/local/lib/brutefir
```

4. open-media-drc (DRC engine + web UIs + DAC hotplug) — the classical CMake build:

```
git clone --recursive https://github.com/delleceste/open-media-drc ~/DRC/open-media-drc
cd ~/DRC/open-media-drc
cp host.cmake.sample host.cmake # AUDIO_USER (defaults to the invoking user),
$EDITOR host.cmake # GEOMETRY, GEOMETRIES, OMDRC_SITE_DATA_DIRS,
# MUSIC_DIR, VIDEO_DIR, OMDRC_API_KEY

mkdir build && cd build
cmake .. -C ../host.cmake
make
sudo make install # -> $PREFIX (default /usr/local)
```

host.cmake is read only by -C, and only for cache entries that do not exist yet. A plain `cmake ..` configures the whole project from the built-in defaults, and adding `-C` afterwards cannot repair that build directory — CMake skips an initial-cache assignment whose entry is already set. The symptom is silent: everything configures, with the wrong values. The project therefore detects it, because `host.cmake` sets a marker the check looks for:

```
CMake Warning at CMakeLists.txt:50 (message):
  host.cmake exists but this build directory was not initialised from it, so
  the built-in defaults are in effect. Adding -C now will not help; start a
  fresh build directory:

  rm -rf build && mkdir build && cd build && cmake -C ../host.cmake ..
```

A `host.cmake` copied from a version of the sample that predates the marker will trip this warning; add the one line the sample carries near the top.

`host.cmake` (the successor to the old `config.env`) is the single source of box-specific values; CMake renders every config from it and installs the DRC engine (`drc.sh` behind the `omdrc / omdrc-status` wrappers), the site data (brutefir configs + impulse-response filters for `GEOMETRY` and every set listed in `GEOMETRIES`, each looked up along `OMDRC_SITE_DATA_DIRS` and reported at configure time), both web UIs (`omdrcctrl :9090` as a **system** service running as the audio user; `omdrcvideo :9080` as a **user** service, since it drives the desktop-session mpv), and the DAC-hotplug glue. The install prints the OS-specific enable steps and the one or two files that must be copied into `/etc` (next section).

The older `./install.sh` rendered the `*.in` templates in place and ran everything straight from the checkout (`git pull = update`). That run-from-repo mode still works but is superseded by the CMake install, which now covers the whole DRC stack — engine, DAC-hotplug glue, both web UIs, and the MPD + upmpdcli renderer configs/units. `install.sh` remains only for the desktop glue not yet in CMake: the browser-nodrc .desktop launcher entries and the Linux `snd-aloop` module-load. (The video mpv-idle autostart entry moved to CMake: `make install` puts it under `$PREFIX/share/omdrcvideo/autostart/` on both OSes, `make user-install` links it into `~/.config/autostart/`.)

5. BruteFIR defaults — BruteFIR reads float precision, partition size and the I/O devices from `~/.config/BruteFIR/brutefir_defaults.conf`. The per-rate configs deliberately leave their input/output blocks empty and inherit devices from this file, so it **must** be deployed:

```
mkdir -p ~/.config/BruteFIR
cp brutefir_defaults.linux.conf ~/.config/BruteFIR/brutefir_defaults.conf # Linux/ALSA
cp brutefir_defaults.conf      ~/.config/BruteFIR/brutefir_defaults.conf # FreeBSD/OSS
```

If this file is missing, BruteFIR (≥ 1.1) silently auto-generates a broken fallback `~/.brutefir_defaults`, every start fails with *“Parse error: path not set ... module ‘file’”*, and `drc.sh` rolls back to direct output — so DRC never comes up at boot.

3.3 Files that must live in /etc

The CMake install copies everything into `$PREFIX` (default `/usr/local`); the update path is `git pull` followed by `cmake --build build && sudo cmake --install build`, not an in-place edit of the checkout. Two files are read *before* `$PREFIX` is in the relevant search path, so they are handled specially:

- **Linux udev rule** — udev only scans `/etc/udev/rules.d` and `/usr/lib/udev`, never `/usr/local/lib/udev`. The install places `99-usb-audio-drc.rules` under `$PREFIX/lib/udev/rules.d` and prints the one-line copy into `/etc/udev/rules.d` (then `udevadm control --reload`).
- **MPD /etc drop-in** — overriding the distro mpd unit’s `User=mpd` requires a real file in `/etc/systemd/system/mpd.service.d/` (a drop-in in `/etc` beats one in `/usr/lib`; see the caveat below).

Everything else stays under `$PREFIX` and needs no `/etc` copy: the systemd units (`$PREFIX/lib/systemd/{system,user}`, which systemd *does* scan), the FreeBSD `rc.d` scripts and devd rule (`$PREFIX/etc/{rc.d,devd}`, scanned natively), `drc.sh`, the filters and configs. The BruteFIR defaults still go to `~/.config/BruteFIR/` (below).

3.4 Linux specifics

3.4.1 The MPD User= drop-in caveat (Arch)

The Arch mpd package ships a systemd drop-in (`/usr/lib/systemd/system/mpd.service.d/00-arch.conf`) setting `User=mpd`. Drop-ins always apply *on top of* the main unit, so a full unit override at `/etc/systemd/system/mpd.service` **cannot** override that `User=` — it silently loses. The repo therefore ships a **counter-drop-in** (`etc/systemd/system/mpd.service.d/open-media-drc.conf`, rendered by `install.sh`) that sets `User=@AUDIO_USER@` and the repo config path; a drop-in in `/etc` beats one in `/usr/lib`. It must be a **real file copied into /etc, not a symlink** (early-boot parse, see above). Do not create a full `/etc/systemd/system/mpd.service`.

3.4.2 USB DAC hotplug (udev + systemd)

File	Installed to	Purpose
99-usb-audio-drc.rules	/etc/udev/rules.d/	Triggers the service on DAC plug/unplug
etc/systemd/system/drc-usb-audio.service	/etc/systemd/system/	Starts/stops DRC
mpd.service.d/open-media-drc.conf	/etc/systemd/system/mpd.service.d/	MPD user/config drop-in

The udev rule matches any USB sound-card control device and pulls in `drc-usb-audio.service` (Type=oneshot, RemainAfterExit=yes so the several controlC* events of one plug never start duplicate BruteFIR instances). ExecStart is `drc.sh restore` after a 1 s settle; ExecStop is `drc.sh stop`. Because udev synthesizes ADD events for already-present devices at boot, the same service covers boot and hotplug.

Manual control: `sudo systemctl start|stop drc-usb-audio.service`, `journalctl -fu drc-usb-audio.service`.

3.5 FreeBSD installation and lifecycle

This section is the authoritative FreeBSD installation and init-system reference. It covers every FreeBSD init or devd artifact owned by this repository. Port templates and their rendered copies are one logical script, so they appear in one row rather than as duplicates.

3.5.1 Installation procedure

1. Install runtime packages and load cuse.

```
pkg install bash brutefir virtual_oss musicpd mpc
sysrc kld_list+="cuse"
kldload cuse
```

Install the optional controller/video Python dependencies and renderer packages for the features actually used. The login audio user must be a member of the groups that grant access to sound and USB devices. BruteFIR must never be started as root: an interactive `drc.sh` could not stop a root-owned instance.

2. Render or install the project.

For a run-from-repository installation, set `AUDIO_USER`, `AUDIO_HOME`, `PREFIX`, and the media paths in `config.env`, then run:

```
./install.sh
```

For the package layout, use the FreeBSD port under `freebsd/audio/open-media-drc`. The installed `omdrc_audio` script points at `/usr/local/libexec/omdrc/drc.sh`; the run-from-repository script points at this checkout.

The early-boot files must be regular copies in the system paths, not symlinks into a possibly separate /home filesystem:

```
install -m 755 etc/rc.d/omdrc_audio /usr/local/etc/rc.d/omdrc_audio
install -m 644 etc/devd/omdrc-audio.conf /usr/local/etc/devd/omdrc-audio.conf
sh scripts/prepare-musicpd-rc-conf-dir.sh \
  /usr/local/etc/rc.conf.d/musicpd
install -m 644 etc/rc.conf.d/musicpd/omdrc_audio \
  /usr/local/etc/rc.conf.d/musicpd/omdrc_audio
```

FreeBSD's rc.subr permits rc.conf.d/musicpd to be either a single file or a directory. The preparation helper makes the directory form. If the path was already a file, it is moved unchanged, with its mode preserved, to musicpd/00-local.conf; then the independent omdrc_audio fragment is installed beside it. The helper is idempotent. The direct Make installer and CMake installer call it automatically, while the FreeBSD package performs the same conversion in its PRE-INSTALL script before package extraction. This avoids both an upgrade failure from mkdir on a file and the loss or rewriting of an administrator's existing MPD settings.

Copy the other enabled rc.d scripts from the table below in the same way. After an update, refresh these copies. Runtime configurations may remain symlinked to the checkout once their containing filesystem is available.

Remove obsolete lifecycle files after the new service is installed and tested:

```
# Historical names; they must not coexist with omdrc_audio.
# /usr/local/etc/rc.d/{drc_usb_audio,brutefir_drc,omdrc_sndlink}
# /usr/local/etc/devd/omdrc-sndlink.conf
# /usr/local/libexec/omdrc-hotplug
```

3. Install the user-owned BruteFIR defaults and MPD configuration.

```
install -d -o AUDIO_USER -g AUDIO_GROUP /home/AUDIO_USER/.config/BruteFIR
install -m 644 brutefir_defaults.conf \
  /home/AUDIO_USER/.config/BruteFIR/brutefir_defaults.conf
```

Merge the three named outputs from mpd/musicpd.conf.in: OKT0-DAC, DRC-native, and DRC-resamp. All FreeBSD physical output paths must use /dev/dsp.dac; do not substitute a numbered /dev/dsp0.

4. Configure the lifecycle in rc.conf.

A core installation with the controller and renderer restore service uses:

```
musicpd_enable="YES"
musicpd_config="/home/giacomo/open-media-drc/mpd/musicpd.conf"

omdrc_audio_enable="YES"
omdrc_audio_user="giacomo"
omdrc_audio_dac="0x152a:0x88c5"      # strongly recommended with >1 card
omdrc_audio_capture="ESI U24XL"     # omit when CD input is unused
omdrc_audio_capture_recsrc="auto"

omdrc_renderer_enable="YES"
upmpdcli_enable="NO"
qobuzconnect2mpd_enable="NO"
qobuzconnect2mpd_user="giacomo"
qobuzconnect2mpd_group="giacomo"
qobuzconnect2mpd_homedir="/var/db/qobuzconnect2mpd"

omdrcctrl_enable="YES"
omdrcctrl_user="giacomo"
omdrcvideo_enable="YES"
omdrcvideo_user="giacomo"
```

Remove any numbered default-device assignment such as `hw.snd.default_unit=0` from `/etc/sysctl.conf`. The global OID survives a re-attach, but its **value is a pcm unit number**, so only the role resolver can know the correct value. Other genuinely global sound tunables remain in `/etc/sysctl.conf`.

For CD input, also set `omdrc_cdin_enable="YES"` and `omdrc_cdin_user="giacomo"`. The detailed CD knobs are documented in section .

These are all project lifecycle key families:

Key family	Purpose
musicpd_enable, musicpd_config	MPD boot and configuration
omdrc_audio_enable, omdrc_audio_user, omdrc_audio_drcsh, omdrc_audio_statussh	master audio lifecycle and user boundary
omdrc_audio_dac, omdrc_audio_capture	stable card identities
omdrc_audio_dac_sysctls, omdrc_audio_capture_sysctls, omdrc_audio_capture_recsrc	per-role settings reapplied after every attach
omdrc_audio_rundir, omdrc_audio_lockfile, omdrc_audio_statefile	root boot-lifetime device transaction state
omdrc_cdin_*	optional CD bridge, fully listed in section
omdrc_renderer_enable, omdrc_renderer_prefix, omdrc_renderer_script, omdrc_renderer_statedir	restore the last selected renderer
upmpdcli_enable, upmpdcli_user, upmpdcli_homedir, upmpdcli_config, upmpdcli_pidfile, upmpdcli_logfile, upmpdcli_flags	UPnP renderer worker

Key family	Purpose
omdrctl_enable, omdrcctl_user, omdrctl_env, omdrcctl_pidfile, omdrctl_logfile	web controller
omdrvideo_enable, omdrcvideo_user, omdrvideo_env, omdrcvideo_pidfile, omdrvideo_logfile	video web remote

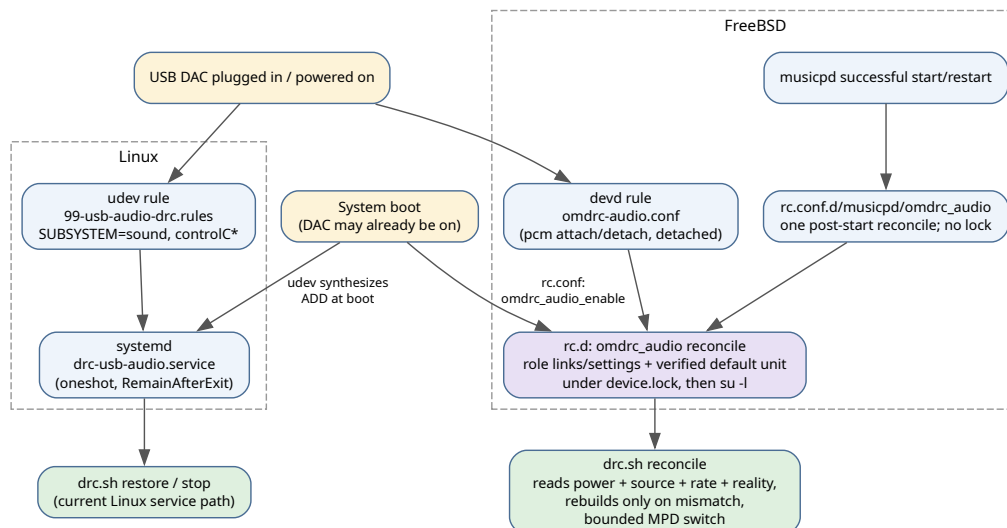
The old `drc_usb_audio_*`, `brutefir_drc_*`, and `omdrc_sndlink_*` families are accepted only as one-release migration fallbacks by `omdrc_audio`. Copy their local values to the new keys and remove them. Enabling an old copied script creates a second lifecycle owner and is unsupported.

5. Validate ordering and activate.

```
rcorder /etc/rc.d/* /usr/local/etc/rc.d/* | \
  egrep 'devd$|omdrc_audio$|musicpd$|omdrc_cdin$|omdrc_renderer$'
service devd restart
service omdrc_audio roles
service omdrc_audio status
service omdrc_audio reconcile
sysctl hw.snd.default_unit      # must equal the pcm unit reported as dac
mpc outputs                    # desired output enabled after musicpd starts
```

`omdrc_audio` requires `devd`, so the cold-plug scan runs after `devd` is listening. A card that finishes attaching after the scan produces a pcm event. The service deliberately does not require MPD: a slow MPD must not prevent the physical DRC chain from becoming healthy.

3.5.2 Complete FreeBSD init and devd inventory



Linux's current udev/systemd edge and the redesigned FreeBSD devd/rc.d level reconciliation.

Script/configuration	rcorder relation or event	Function	Enable directly?
musicpd	REQUIRE: mixer LOGIN avahi_daemon	Starts MPD with the repository's FreeBSD configuration	Yes

Script/configuration	rcorder relation or event	Function	Enable directly?
rc.conf.d/musicpd/omdrc_audio	successful musicpd start_postcmd	Issues one bounded audio reconcile after MPD is actually available	No; sourced by musicpd
omdrc_audio	REQUIRE: FILESYSTEMS devd; shutdown	Single owner of card roles and DRC lifecycle	Yes
omdrc_cdin	REQUIRE: omdrc_audio; shutdown	Optional continuous S/PDIF capture bridge	Yes, only with CD input
omdrc_renderer	REQUIRE: NETWORKING FILESYSTEMS musicpd; shutdown	Restores whichever renderer the UI last selected	Yes
upmpdcli	REQUIRE: NETWORKING FILESYSTEMS musicpd; shutdown	UPnP/OpenHome worker controlled by omdrc_renderer	No when renderer restore is used
omdrcctrl	REQUIRE: NETWORKING LOGIN; shutdown	Starts the web controller as the audio user	Yes when installed
omdrcvideo	REQUIRE: NETWORKING LOGIN; shutdown	Starts the video web remote; it does not start mpv	Yes when installed
omdrc-audio.conf	devd pcm[0-9]+ attach and detach	Detaches one level-triggered omdrc_audio reconcile request	Installed in devd; no rcvar

3.5.2.1 musicpd

The project musicpd script selects the rendered musicpd_config, asks rc.subr to derive the pidfile from that file, and launches the FreeBSD musicpd binary. MPD drops to the user/group declared inside its configuration. It owns no DRC transition and no project lock. The dependency token is consistently musicpd: both omdrc_renderer and upmpdcli require the name this script actually provides.

3.5.2.2 rc.conf.d/musicpd/omdrc_audio

This is a service-specific rc.subr configuration fragment, not another daemon and not another init service. It sets musicpd's start_postcmd to a small function that runs only after MPD has started successfully:

```
omdrc_musicpd_poststart()
{
    checkyesno omdrc_audio_enable 2>/dev/null || return 0
    /usr/sbin/service omdrc_audio reconcile ||
        warn "musicpd: omdrc_audio reconcile failed; retry it manually"
    return 0
}
```

All supported installation paths install this fragment: the FreeBSD CMake branch, the direct Make target, and the port/package. This is required because omdrc_audio normally runs before musicpd: the physical chain can be healthy while its bounded MPD selection remains pending. The successful

MPD start is the factual readiness event that triggers one later reconcile. The hook itself has no lock, wait, daemon, or reverse call to musicpd.

An existing single-file `rc.conf.d/musicpd` cannot coexist at the same path with the fragment directory. Before installing the hook, it is therefore migrated to `rc.conf.d/musicpd/00-local.conf`. `rc.subr` sources regular files from the directory in lexical order, so local settings remain active and the hook is added without editing them.

The trigger exists because `omdrc_audio` deliberately does **not** require or wait for MPD. At early boot it can construct and verify `virtual_oss` and `BruteFIR` first; the final bounded `mpc` output switch then reports pending if MPD is still unavailable. A successful MPD start is the earliest factual readiness signal, so the hook retries once at that event rather than guessing with a delay, polling forever, or adding a readiness gate to the physical chain. The same hook runs after a manual service `musicpd` restart, which is another point at which MPD may have forgotten or disabled its outputs.

The edge is strictly one-way:

```
musicpd successful start
-> service omdrc_audio reconcile
-> roles transaction (device.lock, then release)
-> drc.sh reconcile (drc.lock, bounded mpc)
```

No `omdrc_audio` path starts or restarts `musicpd`, so there is no recursive service cycle. The fragment takes no lock and creates no background process. The ordinary reconcile path supplies all serialization and deadlines. A reconcile failure is reported but the hook returns success: MPD is already running and must not be falsely reported as failed because an optional audio route needs later operator attention. When `omdrc_audio_enable` is not enabled, the hook is a no-op, preserving the master switch.

3.5.2.3 omdrc_audio

`omdrc_audio` replaces the former three-service/helper chain. Its rcvar, `omdrc_audio_enable`, is the only master switch. Its verbs are:

Verb	Meaning
start	boot cold-plug role pass, then full reconcile
roles	root-only role links/settings transaction, no DRC transition
reconcile	role transaction, release device lock, then user DRC reconcile
stop	transient teardown; preserve desired power, rate, design, and source
status	show role resolution and actual chain status

The service absorbs, rather than discards, two essential old responsibilities. From `brutefir_drc` it preserves `su -l`, so `HOME` and the login-class `PATH` are correct and `BruteFIR` remains owned by the same user who runs interactive commands. From `drc_usb_audio` it preserves the master rcvar and

boot/hotplug entry point. It does not preserve the unreliable `/var/run/drc_usb_audio.active` marker; actual processes, configuration paths, rates, nodes, role links, and saved intent are authoritative.

Role publication is an atomic temporary-file rename to `/var/run/omdrc/audio.roles`. Link changes use relative targets:

```
/dev/dsp.dac      -> dspN
/dev/mixer.dac    -> mixerN
/dev/dsp.capture  -> dspM
/dev/mixer.capture -> mixerM
```

A pre-existing non-symlink at a role name is never overwritten. A detach pass removes only project-owned symlinks whose role is now unfilled. An explicit USB ID, optional serial, or description match wins; automatic DAC ranking is reported loudly when several playback candidates make the result a guess. There is no numbered-device fallback.

The same serialized role transaction owns the kernel's bare-OSS default. It sets `hw.snd.default_unit` to `DAC_UNIT` with the absolute base-system `/sbin/sysctl`, reads the value back, logs a change or failure, and makes service `omdrc_audio` status fail visibly when the readback does not match the resolved DAC. This setting matters to applications outside the project that open `/dev/dsp`; project components themselves continue to use `/dev/dsp.dac`. A silent best-effort write is insufficient: on a two-card box, leaving the default at `pcm0` can route an unrelated application into the ESI capture interface even while BruteFIR correctly holds the DAC on `pcm1`.

Do not put `hw.snd.default_unit=N` in `/etc/sysctl.conf`. That file runs near the beginning of `rc`, before USB attach order and roles are known. A fixed number therefore creates two owners and can encode yesterday's enumeration. The role pass is the correct location because it already holds `device.lock`, has selected the card by stable identity, and runs again after each `pcm` attach/detach. It adds no lock and no independent race window.

3.5.2.4 omdrc_cdin

`omdrc_cdin` runs continuously as the configured audio user. It holds the capture role, but opens its playback destination only while audio is present. It prefers `/dev/dsp.play` when DRC is active and can fall back to `/dev/dsp.dac` when DRC is off. The `release extra` command sends `SIGHUP`.

Before replacing `virtual_oss`, `drc.sh` now verifies release through the daemon's line-buffered log. It records the log offset, signals one exact daemon PID, and waits up to a fixed deadline for the new idle/hold acknowledgement and, when playback was held, a new `playback . . . released` record. Failure aborts CUSE teardown. This replaces the old blind 1.5 second sleep, which could destroy a device while the bridge still had it open.

3.5.2.5 omdrc_renderer

`omdrc_renderer` reads `last_renderer` and starts exactly one renderer: `upmpdcli` or `qobuzconnect2mpd`. It retains the selection at shutdown. Both worker `rcvars` remain `N0`; the helper deliberately uses `onstart/onstop`. Enabling a worker independently races the owner and may leave two front-ends driving MPD.

3.5.2.6 upmpdcli

The repository's `upmpdcli` script is a worker service. It supplies the audio user's HOME and a PATH containing `/usr/local/bin`, creates the user-owned pid directory, and captures inherited plugin stderr for the controller's log view. Its REQUIRE token is `musicpd`, not the Linux package name `mpd`.

3.5.2.7 omdrcctrl

`omdrcctrl` runs the Flask controller as the configured non-root user. Its pre-command creates user-writable pid/log directories; its environment sets HOME, PATH, DISPLAY, and optionally the shared `OMDRC_STATE_DIR`. It reads `/var/run/omdrc/audio.roles` without spawning a status command. The DRC panel includes a distinct CD input (44.1 kHz) action.

Its service identity is invariant across callers: `/var/run/omdrcctrl/omdrcctrl.pid` always names the daemon(8) supervisor. The former non-root branch silently selected `${TMPDIR:-/tmp}/omdrcctrl-USER.pid`; consequently an ordinary status probe reported the root-started service as stopped, and `onstart` could attempt a second instance. That branch was removed rather than hidden behind a custom `pgrep` status command. `rc.subr` already avoids a redundant `su` when the caller is the configured service user. `daemon -M 0644` makes the canonical PID readable for diagnostics while the containing directory and service lifecycle remain system-owned.

3.5.2.8 omdrcvideo

`omdrcvideo` starts only the video HTTP/API process. The persistent idle `mpv` belongs to the graphical login session and is not launched by `rc`. The `rc` script sets the non-root user, PATH, pidfile, and logfile and never participates in audio locking.

It follows the same one-service/one-pidfile rule at `/var/run/omdrcvideo/omdrcvideo.pid`. Neither service derives identity from `TMPDIR` or the invoking UID. FreeBSD `rc.d` remains the system lifecycle interface (`sudo service ... start|stop|restart`); a development process must use a distinct port and direct launcher, not masquerade as a second instance of the system service.

3.5.2.9 omdrc-audio.conf

The only project devd configuration matches `pcm[0-9]+` at attach and detach:

```
attach 100 {
    device-name "pcm[0-9]+";
    action "/usr/sbin/daemon -f /usr/sbin/service omdrc_audio reconcile";
};
```

The detach rule has the identical action. Matching `pcm` is a safety boundary. A UAC2 device exposes several USB interfaces but one sound card; matching USB class events caused several lifecycle runs for one plug. A broad USB detach rule could also react to a keyboard or storage device. At the `pcm` event, the kernel has allocated the unit and created its OSS nodes, so role resolution has facts to inspect and does not need a settle sleep.

3.5.2.10 What devd serializes — and what it does not

The current FreeBSD 15.1 source implements a direct action in `sbin/devd/devd.cc::my_system()`. It forks a child which executes `/bin/sh -c command`, while the devd parent waits for that exact child PID with `wait4()`. Consequently, a genuinely synchronous rule such as:

```
action "/usr/sbin/service omdrc_audio reconcile";
```

would block devd until that complete reconcile returned. If a detach arrived while an attach reconcile was running, the detach event would remain pending; after the first action returned, devd would execute the second. The second reconcile would not be lost and the two devd-originated service calls would not overlap under this implementation.

That is an implementation fact, not an API promise. `devd.conf(5)` documents that `action` names a command to execute, but it does not promise synchronous execution, non-overlap, or the present `fork()/wait4()` process model. Correctness must distinguish the FreeBSD 15.1 source behavior from a supported interface which could survive a future devd implementation change.

Our installed action deliberately changes the process lifetime:

```
devd
-> sh -c "daemon -f service omdrc_audio reconcile"
    -> daemon launcher
        -> detached service omdrc_audio reconcile
```

devd waits for its exact shell child. The shell waits for the immediate daemon command, but daemon returns after creating the detached descendant; neither the shell nor devd follows that descendant and waits for the actual reconcile. In FreeBSD, daemon itself always performs the detachment and its `-f` option means **close inherited file descriptors** — it is not a “foreground” or “detach” switch as similarly named tools may use on other systems.

The resulting timing can be:

```
pcm attach A: devd starts daemon A -> reconcile A detaches -> launcher A exits
pcm attach B: devd starts daemon B -> reconcile B detaches -> launcher B exits
               reconcile A and B can now overlap
```

Thus devd still serializes the short top-level launcher commands, but it does not serialize their detached audio workers. This is intentional: a full reconcile may wait on bounded locks, MPD, BruteFIR, virtual_oss/CUSE, and hardware verification. Running it inline would freeze the machine-wide event loop and delay unrelated USB, network, input, storage, and ACPI events.

The device lock makes overlapping workers safe. It is acquired *before* each worker scans pcm state, so a waiter does not later publish a snapshot collected before it waited. For example:

```
A takes device.lock; scans ESI only
OKT0 attaches; B starts and waits
A publishes capture-only state; releases device.lock
B takes device.lock; scans ESI + OKT0; publishes both roles and links
```

The inverse detach case is equally important. Without the lock, an old pre-detach scan could recreate `/dev/dsp.dac` after a newer worker removed it. With the lock, the later waiter scans after acquiring the lock and therefore observes the final device tree. Every request reconstructs complete level state; no attach or detach edge is interpreted directly as an instruction to start or stop DRC.

A detached **rejecting singleton** is not equivalent to either the synchronous model or this queued-lock model. Consider a pidfile wrapper which exits when a worker already exists:

```
A starts and scans: DAC present
DAC detaches
B's top-level devd action runs, but its singleton refuses to start reconcile B
A publishes its older "DAC present" result
```

devd did not lose the detach event: it successfully executed B's launcher. The application-level singleton discarded the reconciliation requested by that event. A correct coalescing singleton would need to set a pending/dirty flag and force A to run another complete scan before exiting. That reintroduces a marker, wake-up protocol, and worker lifecycle while the scan/update transaction still needs serialization. The present lock queues the waiter instead of rejecting it, so the second pass performs a fresh scan.

Authoritative references are the FreeBSD 15.1 `devd.cc` implementation, `devd(8)`, `devd.conf(5)`, `daemon(8)`, and `lockf(1)`.

3.5.3 Locking, concurrency, and bounded waits

Only two locks remain, with a strict non-nesting rule:

Lock	Owner	Protects	Lifetime
<code>/var/run/omdrc/device.lock</code>	<code>root omdrc_audio roles</code>	role discovery, four links, sysctls, recsrc, role publication	one short role transaction
<code>STATE_DIR/drc.lock</code>	<code>audio-user drc.sh</code>	saved intent reads/writes and the physical chain transition	one mutating DRC command

Both FreeBSD calls use `lockf -k -s -t . . . .` -k keeps one inode after release; `lockf(1)` specifically recommends it for concurrent callers because unlink/recreate mode cannot guarantee waiter ordering. File existence does not mean locked. `/var/run` is correct for root boot-lifetime device state; the DRC lock belongs beside persistent user state, and its path is never derived from `TMPDIR`. A desktop session and a boot login shell therefore cannot silently choose different locks.

The order is:

```
devd or rc
-> omdrc_audio roles      [take device.lock; update facts; release]
-> su -l AUDIO_USER
-> drc.sh reconcile       [take drc.lock; converge; release]

musicpd successful start
-> the same omdrc_audio reconcile path (the hook owns no lock)
```

No path takes `drc.lock` and then asks for `device.lock`, and `omdrc_audio` releases `device.lock` before entering the user reconciler. This removes the former three-lock nesting. Several simultaneous pcm events may wait for the short role transaction, then wait for `drc.lock`; the first repairs state and later calls become no-ops. Lock waits have finite timeouts.

The apparent simplification “devd serializes, so delete the locks” is therefore incorrect for four independent reasons:

1. `daemon -f` intentionally releases `devd` before the real worker finishes.
2. Boot rc, the MPD post-start hook, administrators, the UI, and direct `drc.sh` commands are not serialized by `devd`.
3. A rejecting asynchronous singleton can discard the reconcile requested by a busy attach/detach event even though `devd` delivered and executed the top-level action; a queued level rescan must observe the final device tree.
4. The manuals do not guarantee the current `wait4()` behavior.

Removing `daemon` would make long audio waits compromise the entire device subsystem. Replacing the queued lock with debounce plus a dirty marker would add a marker, timer, and worker lifecycle while the scan/update transaction would still need serialization. Combining the locks would hold the short root device transaction across the long audio-user chain transition and recreate cross-privilege lock coupling. The implemented two non-nested locks are thus the simplest safe design, not leftover scaffolding.

The command under FreeBSD’s `lockf file` command is supervised by the `lockf` process. The lock is released when the orchestration command exits; BruteFIR and `virtual_oss` do not become lifecycle-lock owners. Linux uses an explicit flock descriptor and closes it in `daemon` children; the Linux audit is treated separately.

External waits are bounded while `drc.lock` is held. Every `mpc` call goes through a timeout wrapper. A slow or unavailable MPD does not block boot and does not prevent a verified physical chain from becoming active; the pending output selection is logged, and `musicpd`’s successful-start hook supplies the specific next reconcile that retries it. There is no sleep or indefinite MPD readiness loop. Service calls used to restart the CD bridge also have a deadline and use non-interactive `sudo`. BruteFIR startup, exit, `virtual_oss` readiness, DAC warm-up, verification, and CD release all use explicit poll caps.

Syslog is supplementary evidence, not lifecycle state. During the incident, `syslogd` still held a bound but unlinked `/var/run/log` socket; existing connected processes could retain descriptors while new logger calls failed silently. `omdrc_audio` therefore publishes factual role state directly and `drc.sh` appends to persistent `STATE_DIR/drc.log`; neither decides anything from `syslog`. Repair/restart `syslogd` before using log absence as evidence. The project audit found no broad `/var/run` deletion, and FreeBSD `cleanvar` explicitly excludes the `log` and `logpriv` socket names, so the unlink cause remains a separate system issue rather than an attributed project action.

3.5.4 Reconcile state and reboot persistence

The reconciler compares saved intent with physical reality:

- `last_power`: on/off;
- `last_source`: music/cdin;
- `last_arg`: native rate or resampling request and design;

- actual BruteFIR config/process, virtual_oss process/rate/nodes, and DAC role.

A matching chain is a no-op. A missing DAC causes a transient teardown while preserving intent. Desired off remains off. A partial or wrong-rate chain is rebuilt once. The old active marker is not consulted.

The off verb writes `last_power=off` before touching BruteFIR, MPD, CD input, or CUSE. This ordering matters under set -e: a teardown failure must not leave the previous on intent, which would bring DRC back at reboot. stop is different: shutdown and detach use it as a transient teardown and do not change desired power.

restore reads state only after acquiring the same DRC lock, eliminating the old time-of-check/time-of-use window. The explicit UI/CD verb is:

```
drc.sh cdin
```

It records `last_source=cdin` before requesting the 44.1 kHz chain. Reboot therefore restores CD mode at 44.1 kHz. An ordinary rate action is the explicit return-to-music action and writes `last_source=music` before DAC presence and configuration validation or any teardown/startup work. This is write-ahead intent, not a success marker: if the requested music transition fails, the next restore or hotplug reconcile retries music instead of resurrecting the stale CD input at 44.1 kHz. off and transient stop do not change the source. A geometry change in CD mode is refused if the target geometry has no 44100 Hz configuration. The status/UI distinguishes CD input from ordinary 44.1 kHz music.

3.5.5 The verified boot-recursion failure and its removal

The old `omdrc_sndlink` script re-entered itself under `lockf` using `$0`. This was safe under `service(8)`, where `$0` was the absolute script, but false during normal boot: FreeBSD's `/etc/rc` sources each `rc.d` script, so `$0` remained `/etc/rc`. The lock command therefore launched:

```
/bin/sh /etc/rc oneupdate
```

That caused a second rc pass: duplicated host UUID/network/route and service startup, devd already-busy errors, and long wireless timeouts. It was bounded to one extra pass only because the inherited lock marker suppressed another re-entry, but one nested boot was enough to destabilize the machine.

The emergency old-script correction used `rc_service`, because `rc.subr` sets it to the absolute `rc.d` path before sourcing the script. The migration removes that script entirely. `omdrc_audio` retains the safe form for its short locked roles re-entry:

```
/bin/sh "$rc_service" onerole
```

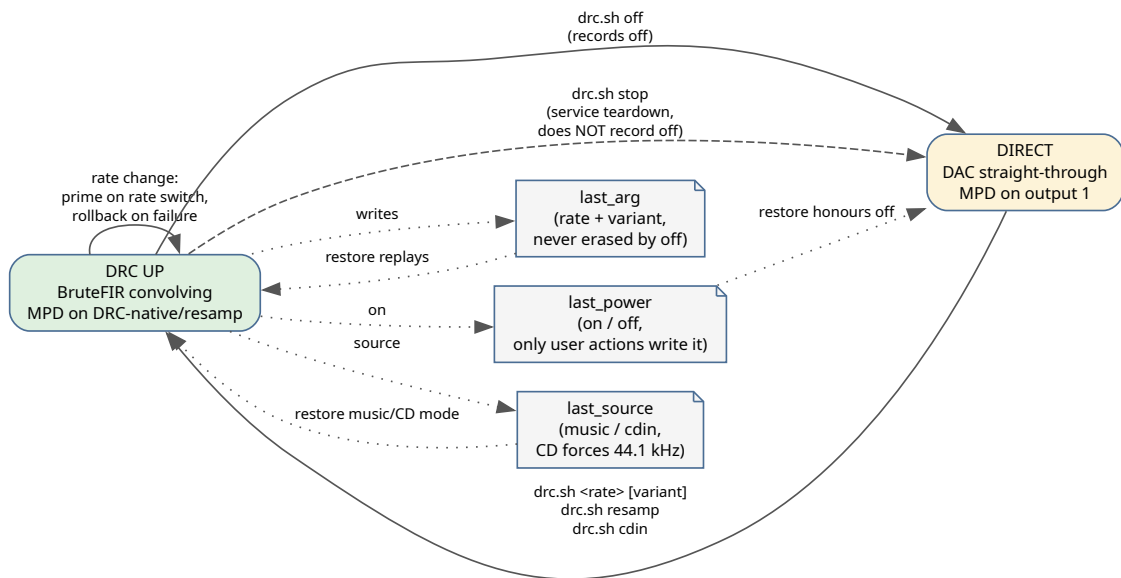
The implementation falls back to the executing script path only outside rc. There is no call to `/etc/rc`, no dirty/debounce recursion, and no helper that can replay boot.

4 Usage: drc.sh, filters, and configuration

4.1 drc.sh — the single control point

```
drc.sh <rate>|resamp|cdin|reconcile|restore|off|stop|status|session [variant]
```

Verb	Effect
<rate>	Start BruteFIR at 44100 / 48000 / 88200 / 96000 / 192000 Hz; restart the loopback at the same rate; switch MPD to DRC - native
resamp	Everything at 192 kHz; switch MPD to DRC - resamp (MPD resamples with soxr)
cdin	Persist CD/S-PDIF source mode and request its required 44100 Hz chain
reconcile	Compare saved power/source/rate with actual processes, config, rate, nodes and DAC role; repair only a mismatch
restore	Re-apply state at shutdown: honour last_power, last_source, and last_arg
off	Stop BruteFIR + loopback; MPD back to direct output; records the off state . The user-facing disable
stop	Same teardown as off but does not record off. Used by service stop paths so a reboot of a <i>running</i> system is restored
status, session	Report actual chain state or the exact persistent restore tuple
geometry, design	Show/list/switch the persistent room geometry or audited filter design
variant	Optional second argument (a config-filename suffix): selects an alternate filter set; superseded by design



drc.sh verbs and the persistent lifecycle state.

State lives in the resolved persistent state directory (git-ignored checkout state in run-from-repo mode, or the configured/installed user state path):

- **last_arg** — the last *active* rate and optional variant (192000, resamp, 192000 <variant>). Written on each successful run, **never erased by off** — turning DRC back on restores the last rate. It records the *desired* state: a failed start never rewrites it, so the next trigger retries the same configuration.
- **last_power** — on or off. Only real user actions write it (a rate run writes on, off writes off; stop leaves it alone).
- **last_source** — music or cdin. CD mode survives reboot and forces 44100 Hz; an ordinary rate selection records the return-to-music intent before fallible validation/transition work, so failure cannot restore stale CD mode on the next boot.
- **last_geometry** — the current geometry override. Design remains part of last_arg, so the full tuple can be restored without guessing.

The configuration's GEOMETRY is the default; `drc.sh geometry <name>` records a runtime override in last_geometry.

What a required rebuild does, in order: stop any running BruteFIR and wait for the DAC to be released; make bounded MPD release requests; verify that CD input has released its output; (FreeBSD) restart `virtual_oss` at the target rate and wait for `/dev/dsp.loop`; prime the DAC if the rate changed; start BruteFIR and **verify it stays up**; make a bounded attempt to enable the matching MPD output; record the state. If BruteFIR cannot come up, `drc.sh rolls back` — stops the loopback and re-enables the direct output — leaving a clean, audible system equivalent to off.

The priming quirk: the OKTO DAC routes silence on the *first* stream opened at a new sample rate; a second open fixes it. On a detected rate change `drc.sh` opens BruteFIR once, tears it down, then starts it for real. (The kernel-level fix for this is the clock-before-alt patch, Appendix ; with it installed, `DAC_PRIME_CYCLES` defaults to 0.)

`drc.sh status` (and `drc-status.sh`) reports the *actual, observed* state — config, loopback/ALSA rate, BruteFIR, MPD output and rate — derived from what is running, not from last_arg.

4.2 MPD outputs

Output	Device	format	Meaning
OKT0-DAC	direct DAC	—	bit-perfect direct path, no DRC
DRC-native	loopback	*:*:*	native-rate DRC; MPD does not convert; you pick the <code>drc.sh</code> rate matching the track
DRC-resamp	loopback	192000:24:2	MPD resamples everything to 192 kHz (soxr); for mixed-rate playlists

::* (rate:bits:channels, asterisk = unenforced) is deliberate: native DRC mode must not force MPD conversion. `drc.sh 192000` and `drc.sh resamp` both use the 192 kHz BruteFIR config but are distinct modes: the web UI shows them as *Flat 192 kHz* vs *Flat auto-resample*. In native mode the sample rate is the routing selector, never the bit depth — BruteFIR processes in floating point.

4.3 The filters/ and configs/ trees

<code>filters/<geometry>/<rate>/{L,R}.raw</code>	raw FLOAT64_LE FIR coefficients
<code>filters/<geometry>/<rate>/@<design>/{L,R}.raw</code>	an immutable A/B design
<code>filters/<geometry>/provenance/<design>.json</code>	hash-bound manifest
<code>filters/<geometry>/analysis/<design>.json</code>	precomputed response traces
<code>filters/<geometry>/rew/</code>	REW-exported source WAVs
<code>configs/<geometry>/brutefir-<rate>[@<design>].conf.in</code>	

These two trees are the *site data*. They need not live in this checkout: CMake resolves them along `OMDRC_SITE_DATA_DIRS` and the design scripts along `OMDRC_SITE_ROOT`, so one room’s measurements can be a separate repository while the engine ships only the generic `flat` set. See [Filter provenance and verification](#).

`drc.sh` builds the config path as `configs/<geometry>/brutefir-<actual_rate><variant>.conf` (for `resamp`, `actual_rate` is 192000) and BruteFIR reads the filter paths from that config — the config is the authoritative link. A variant works only if the matching config file exists; nothing is auto-discovered.

For a new rate or variant, create all the pieces: the `L.raw/R.raw` pair, the config pointing at them, and verify the attenuation: (below).

4.4 Filter generation workflow

This is the low-level route: it produces coefficients but no provenance bundle, so the web UI cannot verify the result and will not show stored measurements beside it. For deployable work use the audited workflow in [Filter provenance and verification](#); what follows is still useful for experiments and is what the audited path drives underneath.

1. Export the corrected impulse responses from REW as WAV (typically 48 kHz).

2. Convert for every rate directory:

```
scripts/REW2raw-all-rates.sh \  
-L filters/120.blue/rew/FLX-trimmed-48k.wav \  
-R filters/120.blue/rew/FRX-trimmed-48k.wav \  
-o filters/120.blue
```

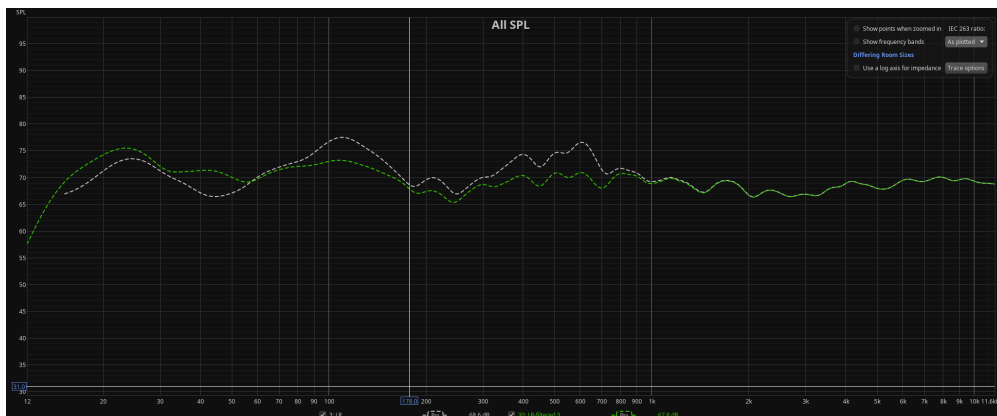
REW2raw.sh (called per rate) resamples with SoX at very high quality (-v -L -s, 64-bit float intermediate) and applies **one deterministic FIR coefficient scale** — $\text{scale} = \text{Fs_source} / \text{Fs_target}$ (Julius O. Smith, *Physical Audio Signal Processing*) — never peak normalisation, which would alter the intended filter gain. A sox.txt log records every command and measured stat.

3. Compute the clipping headroom and set attenuation: in each config:

```
python3 scripts/headroom_calc.py
```

BruteFIR works in float64, so clipping can only happen at the output boundary where the filter has gain > 0 dB. The script FFTs each raw filter, takes the worst-case gain across frequency, adds a safety margin (default 1 dB), and reports the suggested per-pair attenuation (BruteFIR applies one attenuation: per coeff block to both channels, so the louder channel limits). Attenuation in float64 is lossless — the only goal is avoiding clipping.

The measured result of the current filter set (120.blue, v1.5.0):



Amplitude response: corrected vs uncorrected.



Phase response: corrected vs uncorrected.

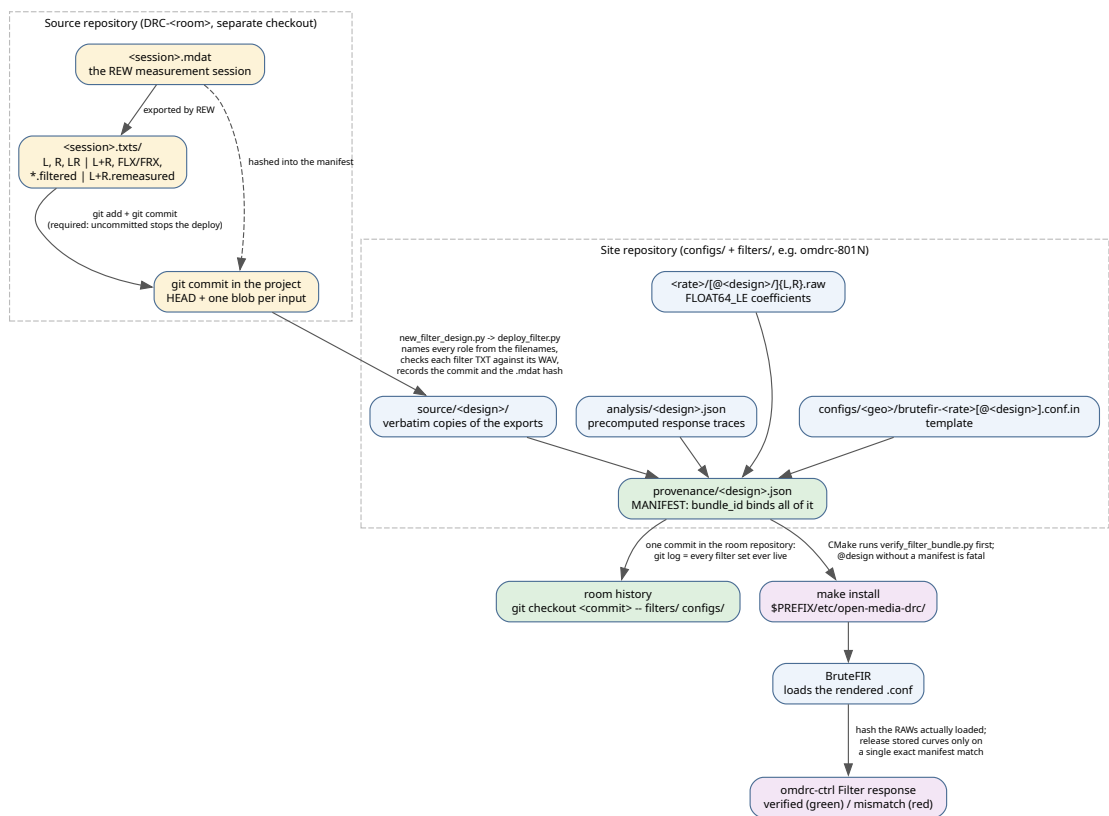
4.5 Browser audio without DRC

While DRC runs, BruteFIR holds the DAC single-open and browsers get silence. Each browser-nodrc/launcher: (1) snapshots `last_power + last_arg`; (2) runs `drc.sh off`; (3) runs the browser in the foreground; (4) restores the exact pre-launch state from an EXIT/INT/TERM trap. It deliberately does *not* use `drc.sh restore` (which would honour the `off` just recorded and leave DRC down). Firefox runs with `--no-remote`; Chrome/Chromium detect an already-running instance with `pgrep` and then just hand over the URL without touching DRC. The rendered .desktop launchers are symlinked into `~/.local/share/applications/`.

5 Filter provenance and verification

A room-correction filter is an empirical artifact: it is only as good as the measurement session it came from, and it is indistinguishable, once converted to a .raw blob of float64 coefficients, from any other blob of the same length. The response page in the web UI shows *stored* room measurements — curves that were computed offline, months earlier, from the original REW exports. Showing them beside a filter that is not the one they describe would be worse than showing nothing, because it looks authoritative.

This chapter describes the machinery that prevents that: a chain of content hashes running from the REW exports in the source repository to the coefficient bytes BruteFIR has actually loaded, with a refusal at every link that cannot be checked.



Provenance chain: each labelled arrow is a content check that must pass before a design is deployable, installable, or shown as verified.

5.1 The three repositories

The workflow spans three trees, deliberately kept apart:

Tree	Holds	Example
Source repository	one geometry's REW projects (.mdat) and the sessions exported from them (<session>.txts/)	./DRC/DRC-120.blue
Site repository	configs/<geometry>/, filters/<geometry>/ — one physical room	omdrc-801N

Tree	Holds	Example
Engine repository	drc.sh, the scripts, CMake, and the generic flat set	open-media-drc

The site data used to live inside the engine checkout. It no longer has to: `configs/<geo>` and `filters/<geo>` are resolved through a *site root*, so personal room measurements can be versioned and deployed independently of the software. Two settings control it, and they are not the same thing as the runtime `OMDRC_SITE_DIR`:

Setting	Read by	Meaning
OMDRC_SITE_DATA_DIRS	CMake	semicolon-separated <i>search path</i> for <code>configs/<geo></code> + <code>filters/<geo></code> ; first match wins
OMDRC_SITE_ROOT	the design scripts	the one checkout they read and write room data in (also <code>--site-root</code>)
OMDRC_SITE_DIR	drc.sh at runtime	the <i>installed</i> <code>\$PREFIX/etc/open-media-drc</code>

Both default to the engine checkout, which is exactly the historical single-repository layout. Set the search path in `host.cmake`:

```
set(OMDRC_SITE_DATA_DIRS "${CMAKE_SOURCE_DIR};$ENV{HOME}/devel/omdrc-801N"
    CACHE STRING "Search path for configs/<geo> + filters/<geo>")
```

At configure time CMake prints which directory supplied each set, so the substitution is never silent:

```
-- core-drc: filter set search path
--   /home/giacomo/devel/open-media-drc (this checkout)
--   /home/giacomo/devel/omdrc-801N
-- 120.blue (default) <- /home/giacomo/devel/omdrc-801N
-- flat <- /home/giacomo/devel/open-media-drc
```

A missing *extra* set is a warning and is skipped; a missing default GEOMETRY is a fatal error, because installing it would record a geometry in `omdrc.conf` with no configs behind it.

5.2 Geometry, design, variant

Three words that are easy to confuse:

- a **geometry** is a physical setup — speaker and listening position, e.g. `120.blue`. It is a directory under `configs/` and `filters/`.
- a **design** is one immutable filter revision inside a geometry, addressed as `@design-id` (e.g. `@rscreen-20260812`) and carrying a provenance manifest. The historical revision has the reserved id `default` and keeps the un-suffixed paths.
- a **variant** is the older mechanism: an arbitrary suffix appended to the config filename. It survives for compatibility, has no manifest, and is therefore always displayed as unverified. New work should use designs.

5.3 The bundle

Everything belonging to one design lives in the site repository under the geometry:

```
filters/<geometry>/
  provenance/<design>.json          the manifest (the commit marker)
  provenance/<design>.source.json    the build recipe (development input)
  analysis/<design>.json            precomputed response traces
  source/<design>/
    L.txt R.txt LR.txt             measured, before correction
    FLX-trimmed.txt FRX-trimmed.txt exported filter responses
    FLX-trimmed-48k.wav FRX-trimmed-48k.wav deployable impulses
    L.filtered.txt R.filtered.txt LR.filtered.txt after correction
  <rate>/{L,R}.raw                 runtime coefficients (design `default`)
  <rate>/@<design>/{L,R}.raw        runtime coefficients (immutable design)
configs/<geometry>/
  brutefir-<rate>.conf.in          template; @REPO_DIR@ -> $SITE_DIR at install
  brutefir-<rate>@<design>.conf.in
```

Sources keep the names they had in the export directory, because those names *are* the role assignment. The copy is deliberate: a deployment must never depend on a mutable sibling checkout or on an absolute path that will not exist on the playback machine.

5.4 What is hashed

The manifest records, for every source export, RAW file, config template and the analysis file: the logical role, relative path, byte size, format, sample rate/count where applicable, and SHA-256. Around that it records where the design came from — the source project's repository, remote, branch, HEAD commit and subject, the repository-relative path of the export directory, a Git blob id per input, and whether everything was committed — together with the REW .mdat those exports were taken from, by name, size, SHA-256 and blob id. It then records the aggregate convention taken from the filenames, parsed REW header metadata (measurement name, date, notes, smoothing, frequency step, timing reference, REW version), the TXT-versus-WAV validation results, per-channel headroom with safety margin and required attenuation, and the exact rate-to-config mapping with the expected BruteFIR format and attenuation.

The `bundle_id` on top is the SHA-256 of a canonical identity object containing a hash of the complete source block — export directory, project, session and every artifact record — plus every source artifact hash, the runtime config and RAW hashes and settings, and the analysis hash. Editing any provenance value the UI displays therefore invalidates the bundle rather than quietly changing a label; that includes the project commit and the .mdat hash.

A hash proves bytes; it cannot say where they came from or bring them back. The commit id does that. It is the only part of the chain that is not self-verifying, which is why the deployment refuses to run until the exports and the .mdat are committed.

5.5 The scripts

All of them are offline. None starts REW, and none opens a .mdat project except the optional auditor below.

Script	Role
<code>new_filter_design.py</code>	the one deployment command: reads one export directory, resolves every role from the file names, checks each filter TXT against its impulse WAV, hashes the .mdat, requires the project commit, reports, asks, publishes, and commits the room repository
<code>remove_filter_design.py</code>	the exact inverse: removes one design completely — manifest, analysis, source copies, every rate's coefficient pair and every config template — manifest first, then records the removal in the room repository. Refuses the reserved default set
<code>deploy_filter.py</code>	the engine underneath: regenerates every requested rate in a temporary directory, validates TXT against WAV, computes headroom, bakes and reads back each config, carries the exports into the analysis file unchanged, and writes the bundle
<code>verify_filter_bundle.py</code>	read-only re-verification of committed bundles: bundle id, source copies, analysis dependencies, configs, exact RAW hashes and headroom. - - no - next for CMake and CI
<code>console_ui.py</code>	shared terminal contract for publication and removal: stages, colours, confirmation, warnings, and the uniform failure line
<code>filter_workflow_next.py</code>	shared operator handoff — prints the exact commit/install/select/verify commands, and names a working directory per step when the site data lives in its own repository
<code>headroom_calc.py</code>	minimum attenuation: per pair from the worst-case FFT gain plus a safety margin; also reports what each config currently specifies
<code>rew_mdat_audit.py</code>	optional archival evidence: audits selected REW project traces via the REW API, comparing TXT responses numerically and final WAV impulses sample-by-sample against the project. Not a deployment dependency
<code>REW2raw.sh</code> , <code>REW2raw-all-rates.sh</code>	the low-level SoX conversion underneath, usable directly for experiments; they produce no provenance bundle

5.6 The workflow

```
# 1. In the source repository: give the exports their imposed names, commit them
#    together with the .mdat they came from.
git -C ../DRC/DRC-120.blue add -- 120.blue.Rscreen.txts 120.blue.Rscreen.mdat
git -C ../DRC/DRC-120.blue commit -m 'Rscreen measurement session'

# 2. In the engine repository: one command, one directory.
export OMDRC_SITE_ROOT=~/devel/omdrc-801N
python3 scripts/new_filter_design.py ../DRC/DRC-120.blue/120.blue.Rscreen.txts

# 3. Re-verify independently, then push the room's history.
python3 scripts/verify_filter_bundle.py --all --require-sources
git -C ~/devel/omdrc-801N push
```

Nothing on the command line says what a file is: L.txt, R.txt, LR.txt (or L+R.txt), FLX-trimmed.txt, FRX-trimmed.txt, the two impulse WAVs, and L.filtered.txt, R.filtered.txt, LR.filtered.txt (or L+R.filtered.txt, or L+R.remeasured.txt when the room was measured again with the DRC running). A missing name, a duplicate spelling or a mismatched aggregate style stops the run in colour before anything is written.

Three checks run before anything is hashed or written, and none of them has an override. Every text export must be **unsmoothed** (* Smoothing: None; one that states no smoothing is refused too, since it cannot be shown to be unsmoothed) — REW's smoothing is baked into the numbers and cannot be undone downstream, while the browser's Smoothing selector is a separate, reversible view. Every text export must come from a **measurement at 48 kHz or below**, checked as *it must not reach past 24 kHz*, because the runtime coefficients are all resampled from one 48 kHz impulse. And each **filter TXT must be the exported response of its WAV**: one integer causal delay and one constant export gain are detected, then the residual magnitude and phase errors must stay inside the declared limits — the only DSP in the pipeline, and what makes the plotted FLX/FRX curve a statement about the bytes BruteFIR will load.

The command reports the eight curves the web remote will plot, the project and session behind them, the two filters with their TXT/WAV residuals and every path it will write, and then asks. --dry-run runs every check, including the SoX conversions, and stops without asking.

The safety controls are intentionally narrow. --yes supplies confirmation but skips no check. --replace-design permits differing bytes under an existing design id but does not permit partial publication. --allow-uncommitted records the source-recoverability gap as clean: false, which remains visible to the verifier and UI. --no-commit gives up the automatic room-history commit but does not alter bundle verification. --site-root selects the site checkout for the scripts; it is separate from CMake's OMDRC_SITE_DATA_DIRS search path.

Publication is a transaction. Without confirmation every check runs in temporary storage and nothing is touched; the dry run also reports which runtime files *would* change. On confirmation the order is fixed: source copies first, then the analysis, then the runtime RAW pairs, and the manifest **last**. The manifest is the commit marker — readers ignore an incomplete deployment until it exists and verifies every preceding hash. Replacing bytes that already exist additionally requires --replace-design, so an accidental overwrite of a deployed design cannot happen silently.

A design owns `filters/<geo>/source/<design>/` and every `filters/<geo>/<rate>/@<design>/`, so after publication those directories hold exactly what the new manifest names — a redeployment that renames its inputs prunes what the previous one left there. Pruning runs after the manifest, so a failure leaves harmless leftovers rather than a manifest naming a deleted file. A bare `filters/<geo>/<rate>/` is shared with the default set and is never touched.

Removal inverts all of this:

```
python3 scripts/remove_filter_design.py --list
python3 scripts/remove_filter_design.py 120.blue@rscreen-20260812
```

It deletes the manifest, the recipe, the analysis file, the source copies, every `@design` coefficient directory and every `brutefir-<rate>@<design>.conf.in`, and nothing else. The manifest goes **first**: while it exists it is the claim that the rest is present, so removing it is what makes the design cease to exist for every reader. The reserved default set is refused, because a geometry falls back to it. The removal becomes one commit in the room repository, and the design stays recoverable from the deployment commit that introduced it.

Step 2 ends with one commit in the room repository, staging `filters/<geometry>` and `configs/<geometry>` and naming the geometry, design, bundle id, project commit, session hash and rates. That commit is the deployment history: `git log` lists every filter set that was ever live, and `git checkout <commit> -- filters/<geo> configs/<geo>` brings any of them back byte for byte. `--no-commit` skips it. Verification never depends on it.

5.7 Deploying the verified identity

Publication changes the site repository, not the installed playback tree. The handoff has four boundaries: the source-project commit makes every export and the `.mdat` retrievable; the site-repository commit records the published bundle; CMake verifies and copies that bundle; and the running BruteFIR config plus exact L/R RAW hashes determine whether the web UI can go green.

On the design machine, `new_filter_design.py` makes the site commit by default. Re-verify those bytes and push the site repository:

```
export OMDRC_SITE_ROOT=/devel/omdrc-801N
python3 scripts/verify_filter_bundle.py --all --require-sources
git -C "$OMDRC_SITE_ROOT" status --short
git -C "$OMDRC_SITE_ROOT" push
```

On the playback machine, pull the site repository. `host.cmake` must include it in `OMDRC_SITE_DATA_DIRS`. Initialise a first build from that file; later deployments can reuse the correctly initialised cache:

```
git -C ~/devel/omdrc-801N pull

# First build for this host
cmake -C host.cmake -S . -B build

# Later deployments may reconfigure the existing cache
cmake -S . -B build

cmake --build build
sudo cmake --install build
```

Configure prints the checkout that supplied each geometry and runs the read-only verifier before anything can be installed. The install contains the RAWs, rendered configs, manifests and analysis JSON; source exports and recipes remain development-only in the site repository.

Restart the panel, select the installed geometry and immutable design, and then verify what is actually running:

```
# FreeBSD; use systemctl restart omdrcctrl on Linux
sudo service omdrcctrl restart
/usr/local/bin/omdrc geometry 120.blue
/usr/local/bin/omdrc design --list
/usr/local/bin/omdrc design @rscreen-20260812
```

Open `http://<box>:9090`, enter **Filter response**, and require the green identity to show the selected geometry/design and the complete `bundle_id` printed during publication and verification. A missing or different identity means the curves must not be trusted. The scripts' **NEXT** block is the host-specific version of this sequence and names the correct working directory for each step in a split engine/site layout.

5.8 Verification at install time

`cmake/core-drc.cmake` runs `verify_filter_bundle.py --require-sources` over every manifest of a geometry *before* installing it, and fails the configure step if a bundle does not verify. It also rejects any `@design` config that has no same-named manifest. A broken bundle therefore stops the build rather than producing an installed system that looks authoritative.

The install copies the manifests and analysis JSON beside the RAWs, and deliberately excludes `rew/`, `source/` and the `.source.json` recipes: their hashes and parsed metadata are already embedded in the installed manifest, so the playback machine needs none of the development material.

5.9 Verification at runtime

The response page never trusts the geometry name or the rate. For every request `omdrc-ctrl`:

1. finds the `.conf` of the **running** BruteFIR process from its argv;
2. parses the coefficient blocks and hashes the exact `.raw` files named there;
3. requires exactly one manifest matching that (relative path, SHA-256, format, attenuation, rate) tuple;
4. verifies the analysis file's own hash and that its recorded input hashes equal the manifest's source artifact hashes;

5. releases the stored exports **unmodified** — no scaling, offset, attenuation subtraction or resampling stands between the analysis file and the browser.

Only then are the stored room measurements released, with the green banner carrying the bundle id; the details panel adds the export directory, the source project and commit, the `.mdat` behind the measurements, every plotted export with its hash and the active L/R RAW hashes.

Anything else is **mismatch** (red): no manifest matches the active bytes, a hash differs, the config attenuation or format differs, or an analysis dependency fails. **No graph is drawn** — not even a diagnostic FFT of the live coefficients, because a calculated curve on a page whose whole promise is *these are REW's numbers* is worse than no curve. A legacy variant, having no manifest, always lands here.

A/B switching obeys the same rule. After `drc .sh` returns, the server re-reads the running process, parses the config actually in use, hashes its RAWs, and reports the new selector as verified only if that runtime identity matches one manifest. A design that starts but fails this check is an assurance failure, not a successful switch.

5.10 What deliberately cannot go green

- A pair whose bytes do not reproduce from the recorded source exports. It must be re-exported and redeployed as a new bundle.
- Any selector without a manifest, whatever its audio quality.
- A design whose configured attenuation is below the computed requirement: the base 120 `.blue` pairs peak at about +1.28 dB and need 2.3 dB including the 1 dB margin, and are configured at 3.0 dB. A design whose filters never exceed unity gain legitimately requires 0.0 dB.

One gap is worth naming explicitly: the manifest pins the config *template*, not the rendered `.conf` that BruteFIR loads, because `@REPO_DIR@` is only substituted at install time. Runtime verification closes this for everything that matters — coefficients, format, attenuation and rate are all re-checked against the bytes in use — but a post-install hand-edit to a rendered config's routing or device settings is outside the chain.

6 Tools

6.1 omdrc-ctrl — the web control panel

A Flask app serving a dark, touch-friendly control panel to any browser on the LAN (default 0.0.0.0:9090). Originally a replacement for KDE Connect's feedback-less "Run command" plugin. Everything is driven by a plain INI file, `commands.conf`; no command is hard-coded.

Widget types — each [section] of `commands.conf` is one command:

- **READ** — runs a shell command, shows its output next to a label, optionally auto-refreshing (`refresh = N` seconds). A READ widget with `details_root` gains a dynamic **Details** button whenever `{details_root}/{output}/README.md` exists — so each filter configuration can carry its own documentation page with images.
- **WRITE** — a labelled button firing a command; green on success, red on failure, optional confirmation dialog (`confirm = yes`).
- **LINK** — opens a URL in a new tab.

Built-in monitoring panels (always present, below the command cards):

- **MPD panel** — the audio-health centrepiece for a headless server: daemon state, playback state and song, the stream MPD reports (rate/bits/ channels), the **DAC feed** (ALSA `hw_params` read from `/proc/asound` on Linux; the `virtual_oss` rate on FreeBSD), the BruteFIR rate, a green/red **SAMPLE RATE MATCH / RESAMPLING** comparison, and a plain-language **bit-perfect verdict**: *Bit-perfect passthrough* (DRC off, all rates equal), *Full-resolution DRC, no resampling* (BruteFIR at native rate), or *Resampling active*.
- **Qobuz Connect panel** — current track via `qobuzconnect2mpd`, with restart button and colour-coded log viewer; plus a renderer switch (`qobuzconnect2mpd` vs `upmpdcli` — never both) driven per-OS (`systemctl --user` on Linux, `sudo service ... onestart/onestop` on FreeBSD).
- **CD input panel** (FreeBSD) — what `omdrc-cdin` is doing, read entirely from its log: an availability LED, the state in one line, the newest failure in another, the [stats] line as chips, and a Start/Stop button. It is the size of its news — one line while no disc is playing, full size while one is (chapter).
- **BruteFIR CPU** — per-process CPU for every `brutefir` instance (matched by `argv[0]`, because on Linux `brutefir` renames its `comm`).
- **Audio Devices** — `/dev/sndstat` on FreeBSD with `fmt 0x...` bitfields decoded to `AFMT_*/PCM_CAP_*` labels; collapsible.
- **Advanced** (FreeBSD only) — `sysctl dev.pcm.<DAC unit>` (resolved from `/dev/dsp.dac`) and `sysctl hw.usb.uaudio` diagnostics.
- **Top CPU** — processes above a configurable threshold.
- **Debug card** — the glitch-detection switch (section).

DRC filter response page — charts the *live* filters loaded by the running BruteFIR: magnitude (dB), delay-compensated wrapped phase, and residual group delay, computed on demand by FFT (NumPy) from the active config's `L.raw/R.raw`. `Chart.js` is vendored, so the page works offline. The filter files are only ever read, never modified.

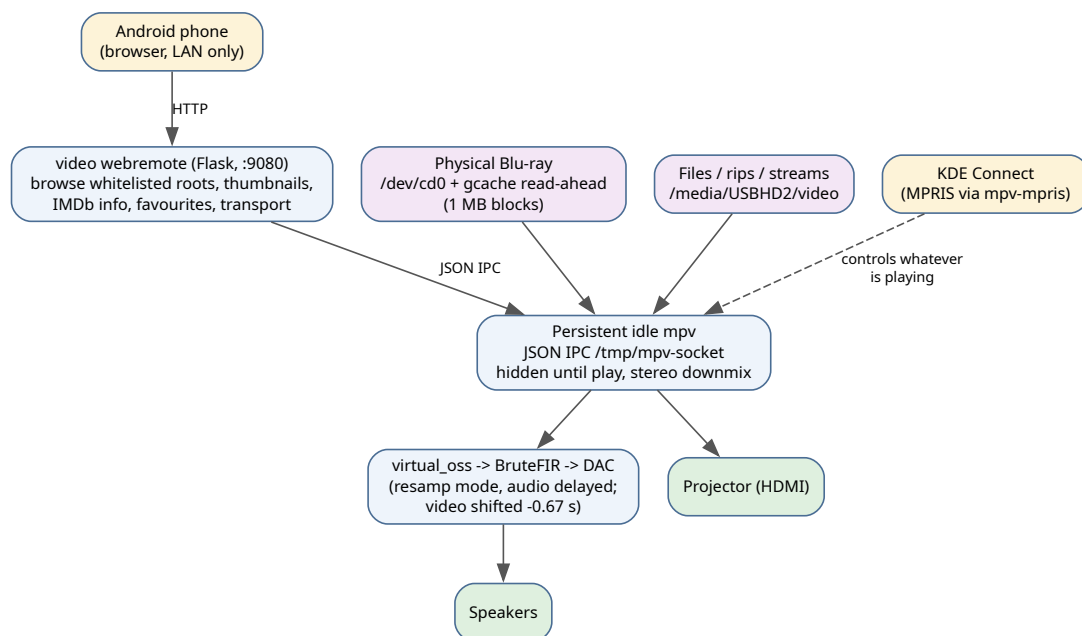
Live spectrum analyzer — an optional card fed by a secondary MPD fifo output (OMDRC Spectrum, raw S32_LE stereo copy of the stream):

- Started/stopped from the page; the MPD FIFO output is enabled only while at least one browser is streaming (Server-Sent Events; multiple clients share one capture thread), force-disabled at startup for crash recovery.
- 24 logarithmic bands from 31.5 Hz, 10 Hz refresh, Music (16384-point) vs Precision (65536-point) FFT windows, VU bars or needles measured over a ~50 ms trailing window, one Floor slider driving graphs and meters.
- **DRC sync:** the tap is pre-DRC, so while BruteFIR runs the display is held back by the measured BruteFIR path delay (filter group delay + partition latency, cached and recomputed only when the active config changes) so it matches the audible sound.

Install: `CMake.cmake . . && sudo cmake --install .` installs system-wide (systemd unit on Linux, rc.d script on FreeBSD); `-DUSER_INSTALL=ON` (Linux only) installs to `~/ .local` with a systemd `--user` unit — configure it *as the target user, without sudo*, and `loginctl enable-linger` for headless boxes. On FreeBSD: `sysrc omdrcctl_enable=YES && service omdrcctl start`; the rc.d script uses `daemon(8)`, drops privileges via the standard `rc.subr ${name}_user`, and keeps its pidfile in a subdirectory of `/var/run` created by `start_precmd` (plain `/var/run/*.pid` would be root-only).

Security: the server executes arbitrary shell commands from `commands.conf` as the service user — trusted LAN only, never a public interface.

6.2 Video: mpv playback + phone web remote



Video playback and control paths.

6.2.1 Playback launchers

- **play-bluray.sh** — physical Blu-ray discs. Kodi cannot read a physical BD on FreeBSD (raw /dev/cd0 wants 2048-byte-aligned reads; the kernel cannot mount UDF 2.50), so mpv + libbluray read the raw device. Because the USB drive only sustains full speed in ~1 MB chunks and raw cd0 has no kernel read-ahead, the script fronts the drive with a **GEOM cache** (gcache create -b 1048576 -s 268435456 bd cd0 -> /dev/cache/bd), and probes bd_list_titles to play the *genuinely longest* title (mpv has no BD menu support; e/E cycle titles at runtime).
- **play-media.sh** — local files, playlists, and network/stream URLs (m3u8, yt-dlp sites); same DRC audio routing, no gcache.

Both source lib/drc-audio.sh, which ensures the chain is in **resamp mode** before playing — necessary because the direct DAC is bit-perfect (bitperfect=1 on the DAC's unit): a 48 kHz movie on a higher-clocked DAC would play ~2x fast. With DRC up, mpv plays to oss//dev/dsp.play and delays the **video** by DRC_VIDEO0_DELAY (default **0.67 s**) to match the audio-path latency; subtitles ride with the picture automatically.

The 0.67 s is derived, not guessed (full derivation in video/AV-SYNC-DELAY.md): the FIR filter's impulse peak sits at sample 96000 of 524288 taps at 192 kHz = **0.500 s group delay** (the coefficient list *is* the impulse response; the peak is when a transient emerges), plus one BruteFIR partition (32768 samples at 192 kHz = **0.171 s**), plus a little virtual_oss buffering.

DVDs are simpler: they mount fine (UDF 1.x) and are low-bitrate, so mpv dvd:// - - dvd-device=/dev/cd0; commercial discs need libdvdcss.

Remote control: the mpv-mpris package auto-loads into every mpv, so KDE Connect's Android *Media control* (and playerctl, Plasma widgets) drive whatever is playing — play/pause/seek/volume/metadata.

6.2.2 The web remote (video/webremote/)

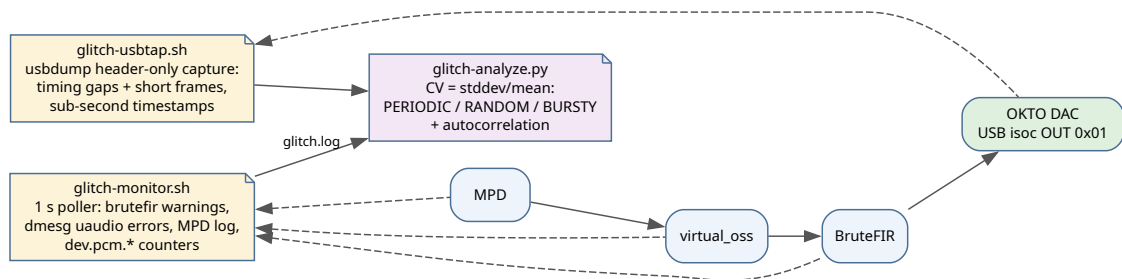
KDE Connect controls what is *already playing*; the web remote is what *starts* a title. A separate Flask app (port 9080, LAN-only, rc.d service omdrcvideo) serving a phone UI to:

- **Browse** the whitelisted media roots (realpath containment — no . . . or symlink escape), with entries classified server-side: folder, Blu-ray rip (BDMV/index.bdmv), DVD rip (VIDEO_TS), or playable file. Grid (poster thumbnails) or compact list.
- **Thumbnails** via ffmpeg frame grabs, disk-cached by path+mtime, with a background prewarm thread and bounded concurrency.
- **IMDb info** — title deduced from the name and, with an OMDb API key, verified and enriched (year, director, cast, plot, rating).
- **Play** on a **persistent idle mpv** over its JSON IPC socket (/tmp/mpv-socket) — hidden until something plays, DRC audio configured once at startup, audio-channels=stereo so 5.1/7.1 sources downmix. Blu-ray rips get the longest-title probe; a **Play Blu-ray disc** button reuses the gcache lifecycle for physical discs, loading into the same mpv.
- **Transport** — seek, +/-10/30 s, play/pause, mute, stop, audio and subtitle track menus; **favourites** pinned to the main page.

The idle mpv is autostarted by the KDE/Plasma session, from `~/.config/autostart/mpv-idle.desktop` (linked to the installed entry by `make user-install`); a `git pull + service omdrcvideo restart` is the whole update path.

6.3 Glitch detection

Implemented for FreeBSD (on Linux the same scripts run but the FreeBSD-specific sources are simply quiet). One global switch — `glitch-debug.sh on|off|status|analyze|usbtap|tail|clear` — also exposed as the Debug card in `omdrc-ctrl`.



The glitch-detection layers and where each taps the chain.

- **glitch-monitor.sh** (always-on, lightweight): polls every second and logs new anomalies from four sources — BruteFIR warnings (missed real-time deadlines), kernel uaudio/USB errors, the MPD log, and any increasing `dev.pcm.*` under/over/err/xrun counter — into a unified `glitch.log`.
- **glitch-usbtab.sh** (definitive, heavier, CLI-only): taps the OKTO's isochronous OUT endpoint 0x01 with `usbdump`, downstream of every software stage. Header-only analysis scales to multi-minute captures; it flags **timing gaps** ($> 2.5\times$ the nominal ~ 4 ms interval) and **short frames** ($SLEN < 0.5\times$ nominal), while the constant $\pm$ -few-samples feedback wobble of asynchronous USB is counted separately and never flagged. Blind spot: a full-length block of zeros (silence insertion) needs payload inspection — use `verify-bitperfect.sh` for that.
- **glitch-analyze.py**: classifies inter-event intervals per stage by the coefficient of variation — $CV \sim 0$ **PERIODIC** (a buffer/clock cycle), $CV \sim 1$ **RANDOM/Poisson** (CPU/scheduling), $CV > 1.5$ **BURSTY** (something waking up) — plus autocorrelation and correlation against DRC rate switches in `drc.log`.

6.4 Bit-perfect verification

`scripts/verify-bitperfect.sh` proves the DAC receives bytes unchanged. Two levels:

1. **Structural** (kernel-certified): with `hw.snd.verbose=2`, `/dev/sndstat` must show the play channel BITPERFECT with the feeder graph exactly `{userland} -> feeder_root -> {hardware}` — any `feeder_rate/feeder_volume/format` node means the kernel is altering bytes. Preconditions: `bitperfect=1` and `play.vchans=0` on the DAC's unit — which is what `omdrc_audio_dac_sysctls` asserts on every attach.
2. **Empirical wire tap** (gold standard): play a deterministic test signal (near-silent ~ -90 dBFS per-sample counter in the low 16 bits, distinct L/R — maximally sensitive to truncation, dither, volume, resampling, channel swap) while capturing the USB isochronous OUT endpoint 0x01 with `usbdump`; decode, align, and byte-compare. The embedded OSS writer aborts loudly if the

kernel coerces format/channels/rate. Verified on this host at 44.1/48/88.2 kHz: hundreds of kB contiguous identical bytes, and on the live MPD both direct (**BIT-PERFECT**) and through `virtual_oss` (**VALUE-EXACT**, 0 slips).

The subtle part is clock domains: a producer must be **flow-controlled by the sink's clock** (blocked writes). MPD is; a free-running test writer is not, and drifts. `virtual_oss` itself is bit-transparent with a flow-controlled producer. The one caveat: BruteFIR bridges the loopback's software clock to the DAC crystal without resampling, so an inaudible one-sample slip occurs every several minutes on the DRC path — values are never altered. The DRC path is *intentionally* not byte-equal (that is the correction); to test its plumbing, use a unit-impulse filter with attenuation 0.

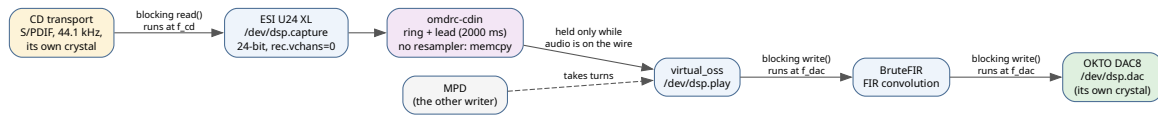
The test asset (`tests/`) is a 44.1 kHz S32 WAV whose PCM payload is byte-identical to the reference raw — MPD cannot play headerless raw.

6.5 scripts/ — helper tools

Script	Purpose
<code>REW2raw.sh</code>	REW WAV -> brutefir raw FLOAT64_LE at a target rate, with the theoretically correct <code>Fs_source/Fs_target</code> coefficient scale (no peak normalisation)
<code>REW2raw-all-rates.sh</code>	Batch: <code>L.raw/R.raw/sox.txt</code> for every numeric rate directory under a filter root; prompts before overwriting unless <code>-y</code>
<code>headroom_calc.py</code>	Minimum attenuation: per config from worst-case FFT gain + safety margin
<code>new_filter_design.py</code>	Publishes a design from one directory of REW exports, and records it in the room's history
<code>remove_filter_design.py</code>	Removes one deployed design completely and records that in the room's history
<code>deploy_filter.py</code>	The offline audit/build engine underneath: rates, validation, analysis, manifest
<code>verify_filter_bundle.py</code>	Read-only re-verification of committed bundles (used by CMake and CI)
<code>console_ui.py</code>	Shared stages, colour, confirmation and failure formatting for design commands
<code>filter_workflow_next.py</code>	Shared operator handoff printed by the commands above
<code>rew_mdat_audit.py</code>	Optional archival audit of REW project traces against exports
<code>verify-bitperfect.sh</code>	The bit-perfect proof tool (above); sources: built-in writer or <code>mpd:OUTPUT</code> ; taps: <code>usb</code> or <code>Loop:/dev/dsp.X</code>
<code>systemd-user-install.sh</code>	Legacy: link + enable a <code>systemd --user drc.service</code> (Linux)

7 CD input: S/PDIF capture into the DRC chain

A CD transport is the second source the chain accepts, and the only one that is not a file: it arrives as a live S/PDIF signal, clocked by the disc, on a USB capture interface. `omdrc-cdin` (FreeBSD only, `cdin/`) bridges it into the same `virtual_oss` entry point MPD uses, so a disc gets exactly the same room correction as everything else.



The CD path. Both ends of the bridge block on their own device, so the ring fill between them is the drift signal; there is no resampler anywhere in it.

It takes the seat `mpv` already takes for video (`video/lib/drc-audio.sh`): a non-MPD writer into the loopback that BruteFIR reads. On Linux the equivalent job is done by `alsaloop(1)`, which is why this is FreeBSD-only code.

7.1 Why a bridge is needed at all

Capture is slaved to the CD's crystal, the DAC runs on its own, and the two differ by a few ppm **forever**. Samples arrive at `f_cd` and must leave at `f_dac`. FreeBSD ships no `alsaloop` equivalent, which is what made this look blocked — but the missing piece was a tool, not a kernel facility: a blocking `read()` runs at the CD's clock, a blocking `write()` runs at the DAC's, so **the ring fill between them is the drift signal**. No OSS clock `ioctl` is needed to measure it.

There is no resampler. The data path is a `memcpy`, so what reaches BruteFIR is bit-identical to what the transport sent — and that is verifiable with `scripts/verify-bitperfect.sh`, which would be meaningless if a resampler sat in the path. Drift is absorbed by the *lead* instead.

7.1.1 The lead is the only number that matters

The lead — how much audio is buffered ahead of the output — is simultaneously three things:

- the **drift margin**: how long before the buffer runs out;
- the **startup delay**: you cannot pre-fill a lead you have not waited for;
- the **transport lag**: every Play/Stop/Skip is heard this much later.

They cannot be tuned separately. The upper bound is arithmetic: `time-to-splice = lead / drift`, so at a pessimistic **50 ppm** a **2000 ms** lead covers about **11 hours** of continuous gapless audio. A disc is at most 80 minutes, therefore **drift cannot cause a discontinuity inside a disc**.

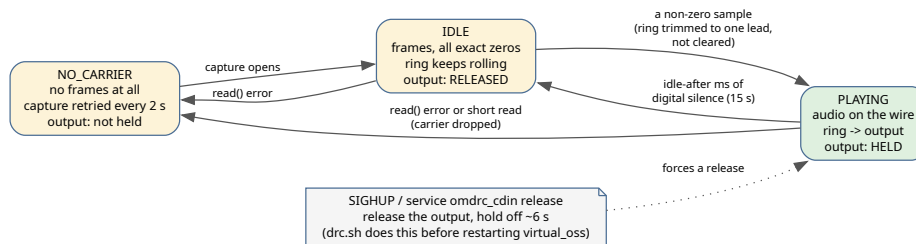
The *lower* bound is not set by drift at all. It is set by transport seeks and USB stalls: some players briefly **drop carrier** across a pregap or index boundary, which is a sub-second input stall that the lead has to absorb. That, not drift, is why the default is 2000 ms rather than the ~50 ms drift alone would need. Below ~250 ms the daemon warns that nothing is left to absorb a seek.

Calibrate it on the real transport with a full disc:

```
omdrc-cdin --in /dev/dsp.capture --out /dev/dsp.play --lead 2000 -d -s 10 \
-l /tmp/cdin.log
```

starves must stay 0 — each one is an audible dropout. After about five minutes the `drift` field reports the measured ppm and the projected headroom, which replaces the 50 ppm assumption with the actual hardware. Step `--lead` down (1500, 1000, 750...) until the first value that produces **any** starve; that is below the floor the transport imposes, so go back up one step and keep a margin. Record the value: it becomes `omdrc_cdin_lead`.

7.2 The three states, and the two very different tenancies



The daemon's state machine. The output device is held only in **PLAYING**; the capture device is held for the whole session.

The daemon is meant to be up all the time — CD player on or off, interface plugged in or not — and that is only safe because of what it does with the *output* device:

State	On the wire	The daemon	/dev/dsp.play
NO_CARRIER	no frames at all	retries the capture device	not held
IDLE	frames, all exact zeros	counts the silence	released
PLAYING	audio	ring -> output	held

The **capture** device is held for the life of a session: it is the interface's own node, nobody else wants it, and it is the only thing that can tell us whether a carrier exists. The **output** device is `virtual_oss`'s client node, and it is taken only for the duration of actual music.

That asymmetry is not politeness. `drc.sh` restarts `virtual_oss` on every rate change, and an open `cuse` client handle at that moment is what wedges the teardown *permanently*: `cuse_server_free()` spins uninterruptibly until every client handle is gone, `SIGKILL` does not touch it, and only a reboot recovers the machine (section). A daemon holding `/dev/dsp.play` around the clock would put a fresh instance of that hazard under every single rate change. For the remaining exposure — a rate change *during* a disc — `drc.sh` asks first:

```
service omdrc_cdin release      # or: pkill -HUP -x omdrc-cdin
```

`SIGHUP` releases the output immediately and refuses to re-acquire it for about six seconds, which covers the stop/restart. Without that hold-off a spinning disc would grab the device again on the very next period. Both `drc.sh off` and its service-teardown stop path request the release before touching `virtual_oss`.

Choosing --idle-after (default 15000 ms) has one failure mode at each end and a wide safe band between them: too short and Red Book's 2 s inter-track pause releases the device mid-disc, so every track change costs a lead to resume; too long and a stopped player keeps holding the chain. --idle-after 0 disables the gate entirely, which is occasionally useful when measuring. Note this is silence *on the wire*: a player that drops carrier instead of sending zeros never reaches the gate at all — the read fails and it lands in NO_CARRIER, which is the state that reopens the device.

Resuming does not lose the first note. The ring keeps rolling through the silence, so an episode begins by *trimming* it to one lead rather than clearing it and waiting for a fresh pre-fill. The music still emerges one lead later, but the period that carried the first sample is still in the buffer.

7.3 What the transport does to the stream

A CD player's S/PDIF output is **always 44.1 kHz**. Transport actions change *what* is sent, never *how fast* — so there is no re-lock, ever:

Action	On the wire	Daemon behaviour
Pause	carrier alive, digital silence (most players)	plays the silence through; lead unchanged
Skip track	brief mute (0.1–1 s), then audio	a short silence, heard one lead later
Fast fwd / rewind	chopped scan snippets or mute, rate unchanged	ordinary audio or ordinary silence
Stop / tray / power	carrier drops	read () stalls or errors, session ends, device reopened

Only the carrier drop matters, and only because the wall clock keeps running while no frames arrive: the lead drains by exactly the dropout's length and never recovers.

7.3.1 Every row of that table is testable without a CD player

Point --in at a **directory** and its *.wav files become the tracks of a disc, played in name order; a single file is a one-track disc. Between tracks the rig emits --gap milliseconds of exact digital silence (default 2000, Red Book's inter-track pause) — which is what the silence gate looks for, so a --gap longer than --idle-after exercises the release/re-acquire cycle with no hardware at all. --transport then scripts the buttons as AT:EVENT pairs, AT being seconds into the stream:

```
omdrc-cdin -i DISC -o /dev/dsp.dac -d -s 5 \
--transport "20:skip,35:pause=4,55:dropout=800,70:seek=+30,105:stop"
```

dropout=N is the important one — it drops the carrier for N milliseconds — and --in-ppm is its counterpart for the *clock*: it offsets the simulated source's pace, which makes the design's central claim testable in seconds instead of the full day real hardware would take. Both failure modes behave as the arithmetic predicts and are worth recognising in the stats:

- **lead exhausted** — starves increments, then in and out converge to the *same* wrong rate. That equality is the backpressure signature: with no buffer left, the DAC is paced by the source instead of by its own clock;

- **ring saturated** — lead pins at the ring capacity and drops climbs at the drift rate; audio is being discarded, one discontinuity per drop.

The rig emulates the CD player's clock, not the disk's seek time, so the disc is prefetched on its own thread (4 s deep) and the paced read is served from RAM — with the read inline, one 3 s stall on an external USB drive drained a 2 s lead and starved playback, which is a property of the rig and not of the design under test. If the medium genuinely cannot sustain realtime the daemon says so, and a schedule that slips more than 100 ms is shifted forward rather than firing every overdue deadline at once (hardware cannot replay frames it missed, and a burst would permanently inflate the lead). Both are counted and surface as `rig_stalls N slips N`, shown only when non-zero.

7.4 Reading the stats line

```
[stats] lead 1635 ms (min 1625, max 1649)  drift +38.7 ppm (+/-387.0),
        ring fills in 46 h  in 44100.206 Hz  out 44088.817 Hz
        frames 4054016/3980288  drops 0 B  starves 0  silence 0%  up 90 s
```

- **Lead is the ring only.** It settles *below* -- lead, because the pre-fill hands the first few hundred ms straight to the output device's own buffer. End-to-end latency is this figure plus that buffer, plus `virtual_oss`'s 200 ms and BruteFIR's filter group delay; the startup delay actually waited is -- lead.
- **drift is measured from the change in lead**, not from the frame counters: those carry each device's constant buffer offset, which at ppm scale would swamp the figure and which cancels in a difference. The +/- is the period quantisation over elapsed time — while it exceeds the estimate, the estimate means nothing. It needs minutes and tightens for hours.
- **in / out are measured from the instant playback began**, and the window restarts at every discontinuity, so a dropout does not leave the cumulative average reading low for the rest of the session.
- **starves** counts events, not periods: one continuous starvation is 1.
- **drift ref dropped (lead jumped)** means the estimate was thrown away and restarted because the lead moved for a reason that is not the clocks. A 3.3 s jump inside a 60 s window once read as +54361 ppm, which is a stall wearing a drift figure's clothes.

7.5 Running it as a service

```
# /etc/rc.conf
omdrc_cdin_enable="YES"
omdrc_audio_capture="ESI U24XL"  # names the ESI; that is what creates
                                # /dev/dsp.capture, which cdin then uses
```

rc.conf variable	Default	Meaning
omdrc_cdin_in	/dev/dsp.capture	capture device (or a WAV file/directory, for the rig); the link comes from <code>omdrc_audio_capture</code>
omdrc_cdin_out	/dev/dsp.play	playback device; /dev/dsp.dac writes the DAC directly, bypassing BruteFIR

rc.conf variable	Default	Meaning
omdrc_cdin_bits	24	source width — the U24 XL's capture endpoint is 24-bit and nothing else (see below)
omdrc_cdin_lead	2000	lead in ms; drift margin, startup delay and transport lag at once
omdrc_cdin_idle_after	15000	digital silence before the output device is released; 0 disables the gate
omdrc_cdin_logfile	/tmp/omdrc-cdin.log	must match log_file in commands.conf's [cdin] — this file is the web card
omdrc_cdin_stats	10	seconds between [stats] lines
omdrc_cdin_user	AUDIO_USER	run user; the same one that owns BruteFIR and MPD, because they take turns on the same devices
omdrc_cdin_flags		anything else: --out-bits, --period, --retry, -v

The rc script is installed by the CMake superproject (cdin/CMakeLists.txt, a subproject of the top-level build). It uses daemon(8), drops privileges via the standard rc.subr \${name}_user, and adds one non-standard verb, release (the SIGHUP above). Started unprivileged with service omdrc_cdin onestart it clears \${name}_user — so rc.subr does not try to su to it — and uses a pidfile under /tmp.

Log lines are a **contract**, not just prose: the web panel parses them, so state <name>: <why> and <device> <path>: <available|unavailable|acquired|released> keep their shape. Availability and holding are separate axes: unavailable means nothing can play and is the red light, while acquired/released is the ordinary rhythm of a daemon doing its job and is never a fault.

7.6 The web panel card

omdrc-ctrl (section) shows a **CD input** card driven entirely by that log, under the Renderer card. It is configured by the [cdin] section of commands.conf, which configures nothing about the daemon itself — only where to read it from, plus whether the buttons exist:

```
[cdin]
enabled = yes
log_file = /tmp/omdrc-cdin.log      # must match omdrc_cdin_logfile
process = omdrc-cdin               # pgrep -x: tells "stopped" from "broken"
service = omdrc_cdin               # what Start/Stop runs
control = yes                      # offer the Start/Stop button at all
refresh = 5
max_events = 20
```

The card is the size of its news, which for a bridge with no disc in the player is one line:

Daemon state	The card
state playing	full size, opened by itself — the only state with anything to watch
idle / no carrier	the status line and an expand chevron; still polling, so a disc reopens it
stopped	the status line and a Start button

Expanding or collapsing by hand sticks until the daemon's own state changes. Expanded, the card adds both device paths and their tenancy, the [stats] line as chips (buffer 1962 ms (min 1955), underruns 0, dropped 0 B, drift +1.2 ppm, fills in 46 h, silence, up), a sentence for anything that has already cost something audible, the raw stats line, and a scrolling event list. Four rules decide what goes where:

- **the LED follows device availability only** — red when an end cannot be opened, because that is the question “can a disc play right now?”. A released output device is the daemon working correctly and never colours anything;
- **failures are kept, health is replaced.** Every error stays in the list, in red, in chronological order, even after the condition clears, and the newest one gets its own red line above the fold. The healthy status line is replaced rather than accumulated. A live status line can never say “the output was missing for ten minutes this morning”, and that is exactly the thing worth saying;
- **starves is the number to surface.** The lead is a margin; an underrun is that margin having run out — a dropout that already happened and that nothing else in the chain will ever mention again;
- **a daemon that is not running is idle, not broken**, however alarming the tail of its log is. The log outlives the process, and nothing is unavailable when nothing is trying to open it. With no daemon and no log at all, the card hides itself.

Three buttons: refresh, **Log** (opens the whole omdrc - cdin log in the Logs card), and **Stop/Start**, which runs the rc service and asks for confirmation before stopping. The exit status of service ... onestart is not trusted — it forks a daemon (8) and returns before the daemon can die on a missing device — so the process is polled with pgrep -x until it agrees. On FreeBSD the button needs a sudoers grant; without one, set control = no rather than leaving a button that can only fail:

```
omdrcctl ALL=(root) NOPASSWD: /usr/sbin/service omdrc_cdin onestart, \
    /usr/sbin/service omdrc_cdin onestop
```

7.7 The ESI U24 XL: what has to be configured, and three traps

The capture interface is an ESI U24 XL (USB Audio Class 1.0, USB 2.0 Full Speed, 32/44.1/48 kHz, 24-bit maximum). Two things the vendor documentation settles, so they are no longer assumptions:

- **it slaves to the incoming S/PDIF automatically** — ESI KB00307EN: when the source is clock master “the U24 XL will receive clock from the source and automatically will be slave”, and there is no manual clock switch. The behaviour this design needs is the only behaviour it has;
- **the sample rate is not auto-detected.** Bit depth and rate must be set to match the incoming signal. Here that is SNDCTL_DSP_SPEED = 44100, which the daemon sets — but it means a non-44.1 source would be captured at the wrong rate rather than refused.

It is class compliant, so `uaudio(4)` drives it natively and there is no vendor driver and no Linux quirk to port (ALSA reaches its input selector through the same generic UAC1 parsing FreeBSD has). **Do not update the interface firmware:** a Linux report has S/PDIF capture working on the original firmware and becoming “completely distorted” after an upgrade.

Everything below was observed on FreeBSD 15.1-RELEASE-p2 and is documented at length in `cdin/ESI-U24XL.md`.

7.7.1 Unit numbers: nothing may depend on them

FreeBSD hands out pcm units in attach order, USB attach order is port order, and on this box the U24 XL sits on a lower-numbered root-hub port than the DAC, so it wins that race at every boot. Whichever card wins, the chain used to address the DAC by unit and nothing else — BruteFIR wrote `/dev/dsp0`, MPD’s direct output was `/dev/dsp0`, `drc.sh` read the clock from `dev.pcm.0.feedback_rate` — so DRC would play into the S/PDIF interface and `cdin` would capture from the DAC.

There is no declarative way to pin the number, which is worth knowing before reaching for one. Unit wiring hints (`hint.pcm.1.at="uaudio0"`) rely on `BUS_HINT_DEVICE_UNIT`, which only `acpi(4)`, `pci(4)` and `isa(4)` implement — `uaudio(4)` does not, so the hint is silently ignored, and worse, `devclass_alloc_unit()` *skips* any unit carrying an `at` hint when numbering an unwired device, so hinting `pcm0` would take unit 0 away from the DAC as well. `hw.snd.default_unit` only selects among the units that already exist; it does not control enumeration. And `devd` is a userland consumer of events the kernel has already acted on: by the time the `ATTACH` notification arrives the unit is allocated and `/dev/dspN` exists, and it has no equivalent of `udev`’s `NAME=/SYMLINK=`.

There is nevertheless one legitimate use of `hw.snd.default_unit`: pointing the unqualified `/dev/dsp` used by unrelated OSS applications at whichever card currently owns the DAC role. Because its value is still a volatile pcm number, `/etc/sysctl.conf` must not set it to a literal 0 or 1. `omdr_audio` writes it inside the locked role transaction after identity-based discovery and verifies the readback:

```
/sbin/sysctl "hw.snd.default_unit=${DAC_UNIT}"
got=$(/sbin/sysctl -n hw.snd.default_unit)
[ "$got" = "$DAC_UNIT" ] || warn "default unit mismatch"
```

This is ancillary to the stable links rather than a replacement for them. BruteFIR, MPD and `cdin` continue to use `/dev/dsp.dac` or `/dev/dsp.capture`; a later kernel or application change to the bare default cannot swap those project roles. Status prints both the discovered DAC unit and the default-unit readback so the distinction is auditable.

So stop trying to fix the number and stop using it. `devfs` accepts symlinks — that is exactly how `devfs.conf`’s `link` directive works, a plain `ln -fs` inside `/dev` run by `/etc/rc.d/devfs` — and the `omdr_audio` service maintains two of them by **role**:

link	is	created
<code>/dev/dsp.dac</code> , <code>/dev/mixer.dac</code>	the DAC everything plays to	always
<code>/dev/dsp.capture</code> , <code>/dev/mixer.capture</code>	the CD/S-PDIF input	only when a capture card is named

Everything opens the DAC by name: BruteFIR's output device, MPD's OKTO-DAC output, `cdin --out`, the mpv launchers, `verify-bitperfect.sh`. A box with one sound card needs no configuration for this to be right, and a box with two stops caring which one enumerated first.

```
sysrc omdrc_audio_enable=YES
sysrc omdrc_audio_capture="ESI U24XL"    # only if you use the CD input
service omdrc_audio status               # prints the roles, exits 1 if unfilled
```

Roles are decided by identity, never by number. The DAC is not simply “the card that is not the capture card”: a plain desktop has onboard HDA on `pcm0` and the USB DAC on `pcm1`, and capability cannot separate them either — an OKTO DAC8 reports (`play/rec`) exactly like a capture interface does (`dev.pcm.N.mode` is a bitmask, `PLAY 0x02 | REC 0x04`, and it reads 7). So an explicit match always wins, given either as a USB id or as a substring of the description `/dev/sndstat` prints:

```
sysrc omdrc_audio_dac="0x152a:0x88c5"      # vendor:product
sysrc omdrc_audio_dac="0x152a:0x88c5:000483" # ...:serial, for two identical DACs
sysrc omdrc_audio_dac="OKTO RESEARCH"      # or just the name
```

With no match configured, playback-capable cards are ranked: a pure-playback USB DAC beats a USB play/record interface, which beats non-USB playback. If several candidates remain, the service reports that it guessed and prints the exact `omdrc_audio_dac` lines that can pin the intended card. The ids come from the card's USB parent, which the service reaches through `dev.pcm.N.%parent -> dev.uaudio.N.%pnpinfo`.

Two triggers, one code path. The `rc.d` service performs the cold-plug pass after `devd` is listening; a later attach/detach comes from `etc/devd/omdrc-audio.conf`, which fires on the **pcm** device rather than a USB interface. Both invoke the same complete reconcile operation:

```
attach 100 {
    device-name "pcm[0-9]+";
    action "/usr/sbin/daemon -f /usr/sbin/service omdrc_audio reconcile";
};
```

By the time the kernel announces `+pcm1 at ... (devaddq(), sys/kern/kern_devctl.c)` the unit is allocated and `/dev/dsp1` exists, so one pass reads the card's identity and links it with nothing to poll for. `devd` is not blocked because `daemon(8)` detaches the request. Matching the USB device instead — system USB, subsystem DEVICE, type ATTACH with vendor/product — fires *before* its `pcm` child exists and would need a retry loop to find it. `reconcile` is a full level rescan, so the same rule covers both roles, detach, a card that moved, and coalesced/reordered events.

The one thing a symlink cannot cover is a sysctl OID. `dev.pcm.<unit>.*` is keyed by the number we are trying to forget, and `drc.sh` reads the DAC's clock from it. That is not a blocker, it is two lines: the unit is read back off the link.

```
t=$(readlink /dev/dsp.dac)                # -> "dsp0"
sysctl -n "dev.pcm.${t#dsp}.feedback_rate"
```

drc.sh has this as `dac_unit()`, the web panel as `_dac_unit()`, and `glitch-usbtab.sh` uses it to find the DAC's `uaudio` parent to tap. Everything that does *not* need the number uses `dac_dev()`, which is `/dev/dsp.dac` when the link exists and `/dev/dsp0` when it does not — so a single-DAC box that never enabled the service, and Linux, where the chain is ALSA, keep working unchanged.

The per-card pcm settings ride with the service, keyed by **role**:

```
omdrc_audio_dac_sysctls="bitperfect=1 play.vchans=0"
omdrc_audio_capture_sysctls="bitperfect=1 rec.vchans=0"
```

`/etc/sysctl.conf` is the wrong home for them for three independent reasons: `dev.pcm.<unit>.*` is keyed by the one thing that is not stable here; `/etc/rc.d/sysctl` runs at rc position 3, long before anything knows which card is which; and a re-attach re-creates the whole `dev.pcm.<unit>.*` tree from driver defaults, discarding anything applied earlier. That last point is why these are applied from the `devd` hook and not only at boot: unplug the DAC and plug it back in, and `bitperfect` is back to 1 before `BruteFIR` reopens it. Most global `hw.snd.*` and `hw.usb.uaudio.*` tunables are not unit-keyed and survive a re-attach, so they stay in `/etc/sysctl.conf`. `hw.snd.default_unit` is the deliberate exception: although the OID is global, its value names a role-dependent pcm unit and belongs to `omdrc_audio`.

Hotplug is now covered rather than refused, and nothing is ever replugged to achieve it — the previous design detached the capture card and USB-reset both so the kernel would renumber them in the wanted order, which cost the capture card its mixer state at every boot and could not be done at all while the chain was playing.

7.7.2 Trap 1: the S/PDIF input is called `pcm2`, and must be selected

The card has two inputs and only one is live at a time. Which one is a **mixer** setting, not a `cdin` setting, and on a fresh boot the card comes up on the *analog RCA* input — so `cdin` records silence from a perfectly healthy CD transport until the recording source is switched:

```
mixer -f /dev/mixer.capture pcm2.recsrc=set      # mixer.capture pairs with dsp.capture
mixer -f /dev/mixer.capture -s                  # print just the active source
```

Only devices flagged `rec` can be a recording source; on this card those are `line` and `pcm2`, which are the two positions of the card's single USB selector unit (`sysctl dev.pcm.<unit>.mixer.selector_0`): `line` = position 1 = analog RCA, `pcm2` = position 2 = digital S/PDIF. They are mutually exclusive, so `set`, `add` and `toggle` all end with exactly one source. The name is not arbitrary: `uaudio` maps `UATE_SPDIF` to `SOUND_MIXER_ALTPCM(uaudio_tt_to_feature[])`, and index 10 of `SOUND_DEVICE_NAMES` is `pcm2`. The `dig1..3` names are only fallbacks for colliding selector pins, and the pre-14 syntax `mixer =rec dig1` no longer exists.

The setting does not survive a reboot, and a replug resets it too — `/etc/rc.d/mixer` only saves state for `mixer0` unless `mixer_enable="YES"` is set, and the U24 XL must be attached at boot for that to see it. So `omdrc_audio` asserts it, on every attach, right after it links the card:

```
omdrc_audio_capture_recsrc="auto"    # the default
```

auto prefers the digital input *by inspection* rather than by hardcoding this card: mixer(8) tags every device that can be a recording source rec and the active one src (printdev()), usr.sbin/mixer/mixer.c), so the service looks for pcm2, then dig1..3, among the sources the card actually offers, and leaves a card that has none of them alone. Name a device (line) to force the analog input instead, or none to keep hands off entirely. The command is idempotent and costs one USB control transfer.

7.7.3 Trap 2: pcm2 = 0.00:0.00 is not a muted capture gain

mixer -f /dev/mixer.capture shows pcm2 at 0.00 next to three devices at 0.75, which reads as a zeroed record level. It is not: there is no gain to raise and raising it does nothing. Sweeping it moves no hardware node, while sweeping line moves them immediately, and there is no software fallback either (sys/dev/sound/pcm/mixer.c applies feeder volume only to SOUND_MIXER_PCM, which is the playback path). The 0.00 is a display artifact: there is no [SOUND_MIXER_PCM2] entry in snd_mixerdefaults[], so it zero-initialises, exactly as line shows 0.75 only because it is in that table at 75. Nothing was measured from the card in either case.

The practical consequence is the correct one for this chain: the capture level on the S/PDIF input is whatever the transport sends, bit for bit, with no gain stage touching the samples.

7.7.4 Trap 3: a “44100 Hz” open that is neither 44.1 kHz nor bit-perfect

This is the expensive one, because every layer reports success. By default FreeBSD puts a **virtual channel** in front of the card. The hardware then runs at dev.pcm.N.rec.vchanrate — 48000, always — and a kernel feeder_rate resamples to whatever the application asked for. SNDCTL_DSP_SPEED returns 44100, SETFMT returns the requested width, every ioctl succeeds, and cdin is satisfied. What actually happens is that the card delivers 44100 frames per second into a stream the kernel believes is 48000:

44100 x 44100/48000 = **40517 Hz** arriving at the application.

cdin then reads 40.5k frames/s and writes 44.1k to the DAC. The lead drains at 3.6k frames/s, so a 2000 ms lead is gone in about 25 seconds, after which the DAC underruns continuously and the music is audibly destroyed. It looks exactly like catastrophic clock drift and it is not drift at all. The only place it is visible:

```
sysctl hw.snd.verbose=2 && cat /dev/sndstat

[dsp1.record.0]: spd 48000 ...                                <-- the hardware
dsp1.record.0[dsp1.virtual_record.0]: spd 44100/48000 ...
... -> feeder_rate(q:4 48000 -> 44100) -> {userland}    <-- the lie
```

The cure is the omdrc_audio_capture_sysctls above — rec.vchans=0 bitperfect=1 — after which the same command prints the whole capture path as {hardware} -> feeder_root(0x00210000) -> {userland}: one memcpy from the USB transfer to the read buffer, at the rate the transport is really running.

7.7.5 The consequence: capture is 24-bit

With the format feeder gone, the card's own width is the only one it accepts, and the U24 XL's capture endpoint offers exactly one — 24-bit S-LE, at either 44100 or 48000 Hz. So `omdrc_cdin_bits` is **24**, and the two ends of the bridge do not share a format: downstream runs S32_LE (`virtual_oss -b 32`, BruteFIR sample: "S32_LE"), and with `bitperfect=1` the kernel's format feeder is gone by design, so the output device does not convert — it refuses the width. `cdin` therefore negotiates and converts itself (`cdin/src/convert.c`): the output is opened at the **source** width first, because no conversion is always true; if the device refuses, a wider one is tried, **never** a narrower one. Widening is left-justification, which for little-endian PCM is pure byte placement (24 → 32 moves bytes and adds a zero) — no arithmetic means no rounding, no dither, nothing to get wrong, so the bit-perfect claim survives it. Narrowing is never offered; it would need truncation or dither, which is what this daemon exists to avoid. `--out-bits N` forces the width instead of negotiating it.

24-bit capture also needed one fix inside `cdin`, and the arithmetic is worth recording because it looks like a bug in the device. `SNDCTL_DSP_SETFRAGMENT` encodes the fragment size as an exponent, so only powers of two can be *expressed*. A 24-bit stereo frame is 6 bytes, so 1024 frames is 6144 bytes = $2^{11} \times 3$ — and no other period rescues it, since $6N = 2^k$ would need 3 to divide a power of two. **Every** period is inexpressible at 24-bit stereo. The fix was to stop treating the fragment as the transfer size: it is the device's *interrupt granularity*, i.e. how much the driver accumulates before waking a blocked `read()`, and the read itself asks for whatever it wants and blocks until that much has arrived. `cdin` now asks for the largest power of two that fits inside the period (4096 bytes for a 6144-byte period) and goes on reading its own 1024-frame periods. The device is woken slightly more often than strictly needed; not one byte of audio changes.

7.7.6 Checklist when the CD input captures silence

0. `service omdrc_audio status` — the U24 XL must hold the capture role with `bitperfect=1 rec.vchans=0`, and the DAC the `dac` role. It also prints the links and the active recording source. If the stream is not silent but *distorted*, and the lead drains to zero in about 25 s, it is Trap 3, not the selector.
1. `ls -l /dev/dsp.capture` — it must exist and point at the U24 XL's unit. If it does not, `omdrc_audio_capture` does not match the card's name.
2. `mixer -f /dev/mixer.capture -s` — must print `pcm2`. If it prints `line`, the card is on the analog input and `omdrc_audio_capture_recsrc` was set to `none` or to `line`.
3. Ignore `pcm2 = 0.00:0.00`; it is cosmetic.
4. Only then look at the transport, the cable and lock.

7.8 Status and what is still owed

The bridge, its state machine, the lazy output open, the width negotiation, the `rc.d` service and the web card are all in place, and the simulated transport has exercised a 12-track 16-bit disc end to end against a bit-perfect `/dev/dsp0` with `starves 0`, `drops 0` and the lead inside a one-period band throughout — including across track gaps, skips, seeks, a 4 s pause and an 800 ms scripted carrier dropout, which cost exactly 800 ms of lead and starved nothing. Three unit-test suites cover the places where a bug would be silent rather than loud: `test_ring` (wrap-around, the drop-oldest policy, every blocking path's wake-up, the trim-to-the-lead), `test_convert` (the widening, pinned by the *value*

relation `src << (dst_bits - src_bits)` plus canary bytes, because a wrong byte index does not crash and does not warn) and `test_gate` (the silence threshold and the fact that one non-zero sample resets the whole run).

What is **not** yet proven is everything on the far side of `/dev/dspN` with a real transport attached:

- **carrier loss:** stop the CD, then unplug the coax. Does `read()` block, short-read, or error? The state machine assumes a short read or an error means `NO_CARRIER`, and that a player which merely *mutes* keeps delivering zeros and is caught by the silence gate. If a stopped player blocks the read forever instead, the daemon needs a read timeout to notice;
- **the real drift:** `omdrc-cdin --in /dev/dsp.capture --out none -d -s 30` reports the capture rate against the host clock; compare it with the OKTO's `feedback_rate`. Their difference is the drift the lead must cover;
- **loopback pacing:** `virtual_oss` runs with `-f /dev/null`, so it owns no hardware clock. Confirm that a writer to `/dev/dsp.play` is throttled at the DAC rate by BruteFIR draining `/dev/dsp.loop`, and check what happens when BruteFIR is *not* running — writes may block forever;
- **the shared-clock patch with two USB audio devices streaming** (section); ideally put the ESI on a different root hub.

Phase 2b — drift resync during the inter-track silence, by padding or trimming with a zero-crossing / 10 ms crossfade splice as the fallback — is not needed inside a disc, since drift cannot cause a discontinuity in 80 minutes, but it is needed for a session that never stops. The seams are marked `TOD0(phase2b)` in the source.

8 FreeBSD peculiarities

A summary of everything FreeBSD-specific, in one place.

8.1 Naming and packages

- **MPD is musicpd**: package `audio/musicpd`, binary `musicpd`, service `musicpd`, client `musicpc` (vs `mpd/mpc` on Linux). `omdrc-ctrl` detects and uses the right client automatically.
- Services are `rc.d` scripts under `/usr/local/etc/rc.d/`, enabled with `sysrc <name>_enable=YES`, run manually with `service <name> onestart/onestop`. Hotplug is `devd`, not `udev`.
- Service names, `rc.d` filenames, `rc.conf` keys, and service-specific hook names use underscores (`omdrc_audio`, `omdrc_audio_enable`); standalone `devd` configuration filenames use hyphens (`omdrc-audio.conf`). The separators are not interchangeable in service, `rcorder`, or `PROVIDE/REQUIRE` tokens.

8.2 OSS instead of ALSA; `virtual_oss` and `cuse`

FreeBSD's native audio API is OSS. The loopback is `virtual_oss` (userland, base system) creating `cuse` character devices: `/dev/dsp.play` (MPD writes) and `/dev/dsp.loop` (BruteFIR reads, synchronized -L mode). The `cuse` kernel module must be loaded (`kld_list` or `etc/rc.d/glue`). BruteFIR's OSS I/O is built in (no ALSA needed); the fork's OSS fixes matter here.

Key `sysctls` for the bit-perfect direct path:

```
bitperfect=1      # first opener's format becomes the hardware format
play.vchans=0     # no virtual-channel mixer/resampler
```

Applied to the DAC's unit by `omdrc_audio` (`omdrc_audio_dac_sysctls`), on every attach — a re-attach rebuilds `dev.pcm.<unit>.*` from driver defaults, so a replug would otherwise silently lose them.

These also mean the DAC is **single-open**: exactly one client at a time (BruteFIR when DRC is on; MPD's direct output or a browser otherwise).

8.3 Known FreeBSD issues and their status

- **OKTO 44.1 kHz-family flicker** (bug #295933): the DAC continuously drops and re-acquires USB streaming lock on 44.1/88.2/176.4/352.8 kHz, while the 48 kHz family is stable and Linux plays everything fine. Root cause: the device has one UAC2 Clock Source **shared** between playback and capture, and `uaudio(4)` lets the vestigial capture side reprogram it to its 48 kHz default under the active playback. Fixed by the shared-clock patch (Appendix).
- **Rate-change cold-open silence**: on the first open after *any* rate change the DAC shows the right rate and streams healthy USB but routes no audio; a second open fixes it. Stock `uaudio` programs the clock *after* selecting the streaming alt-setting (Linux does the opposite). Worked around by `drc.sh` priming; properly fixed by the clock-before-alt patch.

- **virtual_oss livelock** (“155% CPU”, frozen chain): a SNDCTL_DSP_SETRIGGER on a read-only fd could strand the synchronized engine in a wait that nothing wakes. Fixed by the settrigger patch (Appendix).
- **cuse teardown wedge** (bug #296291): stopping virtual_oss could leave it unkillable in D<E state, pinning cuse.ko, reboot required — a kernel refcount leak in cuse_client_open()’s is_closing error path (regression from commit 634e578ac7b0, in 15.1). Kernel fix written (Appendix).
- **pcm unit numbers are attach order, and nothing declarative can change them**: no unit-wiring hint (uaudio(4) does not implement BUS_HINT_DEVICE_UNIT), and no devd rule can pre-empt it (the unit is allocated before the event is delivered). With a second USB audio device present, the DAC can lose /dev/dsp0. Handled by not depending on the number: omdrc_audio keeps /dev/dsp.dac on the right card at boot and on every hotplug, and the few sysctl readers resolve the unit from that link (chapter).
- **A capture open can succeed at the wrong rate**: with a record virtual channel in front of the card the hardware runs at rec.vchanrate (48000) and feeder_rate resamples, while every ioctl reports the rate that was asked for. A 44.1 kHz S/PDIF source then arrives at ~40517 Hz and looks like catastrophic clock drift. rec.vchans=0 plus bitperfect=1 is the cure (chapter).
- **musicpd 100% CPU on HTTP streams**: a libcurl leftover-fd spin — curl registers an event fd into MPD’s I/O loop and never removes it, so the loop burns a core in poll(). **Audio is unaffected**; MPD upstream has triaged it third-party (issues #2244/#2229); the right venue is libcurl.
- **One wire format per attach**: uaudio fixes s32le at attach and pads 16-bit content to 32 bits on the wire, so the DAC panel shows 24 bits on 16-bit tracks. Bit-perfect (zero-padding is lossless), just unlike Linux’s per-stream alt switching. Knobs: hw.usb.uaudio.default_bits/default_rate.

8.4 Video-related FreeBSD constraints

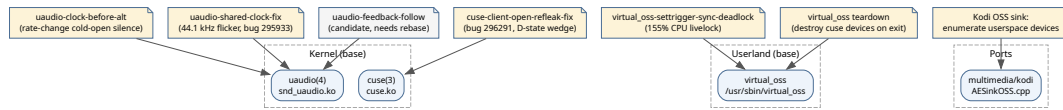
- Physical Blu-ray: no kernel UDF 2.50 mount; raw /dev/cd0 needs sector-aligned reads and has no read-ahead — hence mpv + libbluray + gcache (section). Kodi’s internal player cannot do it.
- Kodi’s OSS sink does not enumerate cuse userspace devices at all — the in-tree Kodi patch fixes that (Appendix).

8.5 Toward a real FreeBSD port

The path from the run-from-repo model to an official audio/open-media-drc port is planned in four phases — upstreaming the patches, splitting engine from site data, the port itself, and submission. The full plan is Appendix .

9 Appendix A — FreeBSD patches in detail

Local fixes kept in-tree while the official FreeBSD fixes are pending. All `/usr/src` patches apply with `-p1` against `releeng/15.1`.



The patched layers at a glance.

Upgrade caveat (applies to every kernel/userland patch below): `freebsd-update` or `make installkernel` overwrites patched binaries with stock ones. After any OS/kernel update, re-apply the patches and rebuild. A `.ko` is ABI-specific to its kernel — never keep prebuilt binaries, always rebuild from the patches.

9.1 uaudio(4) patches (`freebsd-uaudio-patch/`)

Target device: OKTO RESEARCH DAC8 STEREO (USB `0x152a:0x88c5`, Thesycon UAC2 firmware, USB High-Speed). Two patches are applied in order on top of stock `releeng/15.1`; a third is an unbuilt candidate.

9.1.1 1. `uaudio-clock-before-alt.c.patch` — rate-change cold-open silence

Problem. On the first open after *any* sample-rate change the DAC opens the stream, shows the correct rate, streams healthy USB (feedback present, no underruns) — and routes silence. A second open at the same rate fixes it, which is why `drc.sh` had to prime (and why users had to “run `drc.sh` several times”).

Root cause. Stock `uaudio_configure_msg_sub()` starts a stream as: `SET_INTERFACE` to the streaming alt-setting (arming the device), *then* `SET_CUR` the sample rate on the UAC2 Clock Source (possibly a crystal switch, yanked under the armed interface), then start transfers. Linux does the opposite: park at alt 0, program the clock idle, then select the alt. The Thesycon firmware latches its stream configuration at `SET_INTERFACE` time, so FreeBSD’s order arms the stream against the *old* clock. This also explains why the open/close prime works: every close parks at alt 0.

The fix. For UAC2 devices in the `CHAN_OP_START` path: park the streaming interface at alt 0; program the clock while idle (legal — the UAC2 clock lives on the AudioControl interface); on a genuine rate change sleep `hw.usb.uaudio.clock_settle_ms` (default 100 ms, runtime-tunable, clamped to 2000) for crystal relock; then `SET_INTERFACE` and start. UAC1 devices are untouched (their rate control lives on the streaming endpoint).

Status. Built + installed since 2026-07-06; listening reports say different-rate tracks now lock first try. Replaces the host-side `DAC_PRIME_CYCLES` prime for all clients (`drc.sh`, `MPD-direct`, browsers). If a crossing is ever silent, raise `sysctl hw.usb.uaudio.clock_settle_ms`.

9.1.2 2. uaudio-shared-clock-fix.c.patch — the 44.1 kHz flicker (bug #295933)

Problem. Continuous drop/re-acquire of USB streaming lock on the 44.1 kHz rate family; the 48 kHz family is stable; Linux plays everything.

Root cause. The DAC exposes **one UAC2 Clock Source shared between playback and capture**. On async playback, uaudio auto-starts the record channel purely as a jitter-information source — at its own nominal rate (48 kHz default), whose SET_CUR clobbers the shared clock out from under active 44.1 kHz-family playback.

The fix — three cooperating, device-agnostic changes:

- a. **Rate-align the jitter record stream:** before the auto-started record channel starts, set its alt to the one matching the playback rate, so the rec-side SET_CUR becomes a same-value no-op *and* the rec channel's framing stays consistent (valid jitter feedback instead of catastrophically wrong).
- b. **Shared-clock guard:** before any UAC2 SET_CUR to a clock shared by both directions, skip it if the other direction is running at a different rate — the first active stream owns the clock.
- c. **Always submit the explicit-feedback SYNC transfer**, so dev.pcm.<unit>.feedback_rate stays live as a diagnostic (drc.sh's chain-sanity signal) even with capture present.

A guard-only version was audited and rejected: without (a), the rec channel expects 48 kHz framing while the device correctly delivers 44.1 kHz, the jitter estimate pins at its negative clamp, and the play callback strips samples continuously — worse than the flicker. This fix **replaces** the retired VID/PID-gated capture-disable workaround; pcm0 (play/rec) in sndstat is the *expected* state again.

Status. Applied to /usr/src 2026-07-07, builds -Werror-clean alone and on top of patch 1; the combined module was pending install + listening test at the time of writing. Upstream: bug #295933 / PR 2323.

9.1.3 3. uaudio-feedback-follow.c.patch — candidate (unbuilt)

Sketch to make playback follow the device's reported feedback rate smoothly (Linux-style), targeting an occasional tick. Touches the same sync-callback region as fix (c) above, so it **needs rebasing** before any use.

9.1.4 Applying and building

```
cd /usr/src
patch -p1 < uaudio-clock-before-alt.c.patch
patch -p1 < uaudio-shared-clock-fix.c.patch      # applies with offsets; also on pure stock
patch -p1 < Makefile.patch                      # adds -DUSB_DEBUG

cd /usr/src/sys/modules/sound/driver/uaudio
make clean && make
```

Install the module (back up stock first, refresh the backup after every OS update so the revert path matches the running ABI):

```

OBJ=/usr/obj/usr/src/amd64.amd64/sys/modules/sound/driver/uaudio/snd_uaudio.ko
sudo cp -f /boot/kernel/snd_uaudio.ko /boot/kernel/snd_uaudio.ko.orig
sudo service musicpd stop                # release the DAC
sudo cp -f "$OBJ" /boot/kernel/snd_uaudio.ko
sudo kldunload snd_uaudio                 # devd auto-reloads on attach
UG=$(usbconfig | awk '/DAC8STEREO/{print $1}' | tr -d ':')
sudo usbconfig -d "$UG" reset             # clean re-enumeration
sudo sysctl -f /etc/sysctl.conf           # restore buffer_ms baseline
sudo service musicpd start

```

Verify: `grep pcm0 /dev/sndstat` shows (play/rec); `sysctl hw.usb.uaudio.clock_settle_ms` exists; `sysctl dev.pcm.$(readlink /dev/dsp.dac | tr -dc 0-9).feedback_rate` tracks the playback rate. Then run the listening matrix (both crystal directions, same-family changes, MPD-direct mixed-rate queue, browsers) — machine signals cannot verify these fixes; only listening counts. Revert with the `.orig` copy + `kldunload/kldload`.

9.2 virtual_oss / cuse patches (freebsd-virtual-oss-patch/)

Two related bug clusters, both root-caused live on this box.

9.2.1 1. Runtime livelock: virtual_oss-settrigger-sync-deadlock.patch

Symptom. Minutes after a chain (re)start, playback freezes: MPD stops advancing, BruteFIR starves, `virtual_oss` burns 150–200% CPU, both clients stuck in `cuse-cli` waits. Stopping anything from this state walks into the teardown minefield below.

Root cause (captured with `procstat` + `gdb` on the live process): a pair of userland bugs in `usr.sbin/virtual_oss`:

1. `SNDCTL_DSP_SETTRIGGER` ignores the fd's open mode: BruteFIR's OSS layer triggers both directions with one call (harmless on kernel pcm), flipping `tx_enabled = 1` on the read-only `/dev/dsp.loop` fd that can never write.
2. The synchronized-loopback engine wait loops check `tx_enabled` **once, outside the wait**, and the trigger/halt ioctl paths never wake the engine — so once the engine parks waiting for play data from a client that will never write, nothing ever re-evaluates the premise. The arming window is microseconds wide against a 200 ms block cadence — hence intermittent.

The fix (four changes, upstreamable): `SETTRIGGER` honours the open mode (`fflags`); trigger/halt ioctls `atomic_wakeup()` the engine; the sync wait loops re-check `tx_enabled/rx_enabled`; and a new `tx_written` latch so the engine never sleeps waiting for a client that has never written.

9.2.2 2. Teardown deadlock: userland device destroy + kernel reflink

Symptom. Stopping `virtual_oss` intermittently wedges it forever in `D<E / MWCHAN W` (SIGKILL-immune), pins `cuse.ko` (kldunload hangs too), a new `virtual_oss` cannot recreate the devices — reboot required. Every DRC rate change was a reboot risk.

Two layers:

- **Userland:** `virtual_oss` never destroyed its `cuse` devices on exit, so the kernel's `cuse_server_free()` busy-waited in `pause("W", hz)` for client refs that were never released. Patches `virtual_oss-teardown-int.h.patch` + `virtual_oss-teardown-main.c.patch` keep each `cuse_dev_create()` handle and call `cuse_dev_destroy()` on all DSP/WAV/loopback devices before exit. (Upstream committed an equivalent as `0bd5ef6b4363`.)
- **Kernel** (the real fix — bug #296291): a regression from commit `634e578ac7b0` (Nov 2025, in 15.1). In `cuse_client_open()`, the `is_closing / si_drv1==NULL` error path returns with `pcs->refs` incremented, the client left linked in `hcli`, and the destructor never registered — **every open racing into the teardown window leaks one server ref**, and `cuse_server_free()` waits forever. Proven live with a diagnostic `cuse.ko` + `dtrace` + `kgdb` (refs frozen at 4, three leaked clients on `dsp.loop`). Fix: `cuse-client-open-refleak-fix.patch` calls `cuse_client_free(pcc)` on both error paths; committed on branch `fix/cuse-client-open-refleak-296291`, compile-tested, Fixes: `634e578ac7b0`, PR: `296291`. The userland destroy mitigates the normal exit but does nothing for `kill -9` — the kernel leak stays latent without this.
- A diagnostic-only `cuse-teardown-diag.c.patch` logs the stuck refcount every ~5 s from the `pause("W")` loop (pure `printf`; remove once the kernel fix lands).

9.2.3 Applying and building

```
cd /usr/src
patch -p1 < virtual_oss-settrigger-sync-deadlock.patch
patch -p1 < virtual_oss-teardown-int.h.patch
patch -p1 < virtual_oss-teardown-main.c.patch
patch -p1 < cuse-teardown-diag.c.patch          # optional diagnostic

# userland:
cd /usr/src/usr.sbin/virtual_oss && make && sudo make install

# kernel module (diagnostic or refleak fix):
cd /usr/src/sys/modules/cuse && make && sudo make install
```

Reboot first if a wedged `virtual_oss` is pinning `cuse.ko` — a new module cannot load and the devices cannot be recreated until then. Validation: `churn drc.sh <rate> / off` cycles ~20x; `virtual_oss` must stay near-idle CPU and always exit cleanly (no D<E in `ps -o pid,stat,mwchan,cuse.ko` unloadable).

9.3 Kodi OSS sink patch (kodi-virtual-oss-patch/)

Problem. Kodi's audio settings only ever offered the hardware DAC.

`CAESinkOSS::EnumerateDevicesEx()` counts kernel PCM cards via `SNDCTL_SYSINFO`; `cuse` userspace devices are not kernel cards, so `/dev/dsp.play` was never listed — and the settings filler silently *resets* any configured value it cannot match against the enumerated list, so hand-editing `guisettings.xml` never stuck.

The fix. After the kernel-card loop, parse the *“Installed devices from userspace:”* section of `/dev/sndstat` (the only place `cuse` devices are advertised) and add each node, probed exactly like a kernel card (`SNDCTL_ENGINEINFO` for formats/channels/rates, non-blocking open, graceful fallbacks

so a busy device is still listed, dedup against kernel cards). Generic — any userspace OSS device is listed.

Result (verified on multimedia/kodi 22.0a3): *“dsp.play virtual_oss device”* appears in Settings > System > Audio, selecting it persists, and Kodi feeds the DRC chain. `virtual_oss` must be running when Kodi initialises audio.

Apply via the ports tree:

```
sudo cp patch-xbmc_cores_AudioEngine_Sinks_AESinkOSS.cpp \  
    /usr/ports/multimedia/kodi/files/  
cd /usr/ports/multimedia/kodi  
make config && make patch && make build  
sudo make deinstall && sudo make install    # same version -> reinstall is a no-op  
strings /usr/local/lib/kodi/kodi.bin | grep -c "from userspace"    # must print 1
```

Upstreaming: to the FreeBSD port (files/ patch, PR to the maintainer) and/or Kodi upstream (AESinkOSS.cpp PR; pre-empt the “why parse sndstat text” question — cuse devices are unreachable via the mixer ioctls).

10 Appendix B — The FreeBSD port plan

Status: **plan only, nothing applied**. Linux packaging is explicitly out of scope for now (one OS at a time). Source: doc/FREEBSD-PORT-PLAN.md.

10.1 Why the repo cannot be ported as-is

A FreeBSD port installs *identical, immutable* files on every machine, under hier(7) paths, from a versioned release tarball. The run-from-repo model violates that on every axis — deliberately, because it optimizes for a zero-config personal appliance:

1. **Files are rendered per-host:** `install.sh` bakes `config.env` values (@AUDIO_USER@, @AUDIO_HOME@, @REPO_DIR@) into the live files; a package must install the same bytes everywhere and configure at runtime.
2. **The tree is written at runtime:** `drc.sh` keeps `last_arg`, `last_power`, `drc.log` beside itself; `pkg check -s` flags modified packaged files — state must live in `/var/db/`.
3. **Room-specific data is mixed with software:** `configs/120.blue`, `filters/*` (~50 MB) are personal measurement products; a port must ship neutral defaults. `OMDRC_SITE_DATA_DIRS` / `OMDRC_SITE_ROOT` now resolve these in a separate checkout beside the engine.
4. **rc.d scripts shadow other ports:** `etc/rc.d/musicpd` and `upmpdcli` would replace scripts owned by `audio/musicpd` / `net/upmpdcli`; the stock scripts' `rc.conf` knobs must be used instead.
5. **Dependency on a personal BruteFIR fork:** `RUN_DEPENDS` must resolve to ports.
6. **Kernel/userland patches:** a port cannot patch the base system (`uaudio`, `cuse`) and should not carry patches for another port (`virtual_oss`) — these must land upstream first.
7. **Missing packaging basics:** no LICENSE, no tagged releases (the `v*` tag series this manual belongs to is the first step), and the tarball would ship debugging journals and kernel patches.

Run-from-repo does not have to die: the repo becomes a normal upstream that *also* supports `make install PREFIX=... DESTDIR=...`; run-from-repo stays the development/appliance mode and the port is a thin consumer of tagged releases.

10.2 Phase 0 — Upstream the out-of-tree pieces (prerequisite, in flight)

- **uaudio patches** (shared-clock fix, clock-before-alt) to FreeBSD base — in progress: bug 295933 / PR 2323.
- **cuse refleak fix** to base — in progress: bug 296291.
- **virtual_oss SETTRIGGER deadlock patch** to upstream `hselasky/virtual_oss`, so `audio/virtual_oss` inherits it.
- **BruteFIR fork** — options in order of preference: (a) upstream the delta to Anders Torger (upstream is dormant — unlikely); (b) submit the delta as `files/patches` to the existing `audio/brutefir` port (viable if the delta stays small and FreeBSD-relevant); (c) release the fork as its own project + port (`audio/brutefir-omdrc`) — most work, only if b. is refused.

- **kodi-virtual-oss-patch**: not shippable by this port — upstream to Kodi or to multimedia/kodi's files/, or drop from the tarball.

10.3 Phase 1 — Make the repo package-friendly

Worth doing even if the port never happens; it makes the repo usable by anyone, not just this box/room.

- **1.1 Engine / site-data split.** Engine (shipped): drc.sh, drc-status.sh, omdrc-ctrl, video/webremote, browser-nodrc, own rc.d/devd glue, sample MPD/upmpdcli snippets, docs. Site data (not shipped): the configs/ and filters/ measurement products, room README sections, measurement plots — moved to examples/ rooms/ or a separate private overlay checked out beside the engine (OMDRC_SITE_DIR).
- **1.2 FLAT default filters** (first — small and independent): ship a configs/flat/ geometry as default, one conf per rate, using BruteFIR's built-in identity coefficient filename: "dirac pulse" with attenuation: 0.0 — no binary filters needed, and verify-bitperfect.sh can pass through the flat chain as a plumbing self-test. Keep filter_length and I/O identical to the room configs so swapping in real filters is a filename change.
- **1.3 Runtime configuration instead of render-time baking:** drc.sh and friends read a config at runtime (\$OMDRC_CONF -> \${PREFIX}/etc/open-media-drc/omdrc.conf -> <script-dir>/config.env, the last keeping run-from-repo unchanged); rc.d defaults flip to installed paths via the port's SUB_FILES; brutefir confs rendered on the fly to a state-dir tempfile so packaged confs are host-neutral.
- **1.4 State out of the tree:** last_arg, last_power, drc.log to /var/db/omdrc/ in installed mode (beside the script as fallback); /tmp/brutefir.out, /tmp/virtual_oss.pid to the same state/run dir.
- **1.5 Install target** (make install with DESTDIR/PREFIX): scripts to \${PREFIX}/libexec/omdrc/ + a \${PREFIX}/bin/omdrc wrapper; omdrc.conf.sample + flat configs to \${PREFIX}/etc/open-media-drc/; devd conf; MPD/upmpdcli snippets to share/examples/; omdrc-ctrl to share/omdrc-ctrl/; docs to share/doc/open-media-drc/. rc.d scripts for our own services are installed by the port via USE_RC_SUBR.
- **1.6 Housekeeping:** pick a LICENSE (BSD-2-Clause fits the ecosystem); tagged releases whose tarballs exclude freedbsd-* -patch/, investigation journals and site data (git archive + export -ignore); split the README into a user quickstart vs doc/DEVELOPMENT.md.

10.4 Phase 2 — The port itself

audio/open-media-drc (working name): USE_GITHUB=yes, tagged DISTVERSION, NO_BUILD for the shell core, USES=python:run shebangfix for omdrc-ctrl. RUN_DEPENDS: brutefir (per the Phase 0 decision), virtual_oss, musicpd, sox/soxr. OPTIONS_DEFINE: CTRL (web UI: py-flask/Markdown/numpy), VIDEO (webremote: mpv), UPNP (upmpdcli). USE_RC_SUBR for omdrc_audio (plus omdrcctrl with the CTRL option) — **not** musicpd or upmpdcli; a pkg-message documents the rc.conf lines pointing the stock scripts at our configs. @sample entries for every config. Validate with portlint -AC, portclippy, poudriere testport, and pkg check -s after a service run (catches leftover in-tree writes).

10.5 Phase 3 — Submission and maintenance

Submit as a Bugzilla PR (Ports & Packages), MAINTAINER= `delleceste@gmail.com`. Niche integration ports are accepted when clean and maintained — the bar is quality, not popularity. Expect review rounds on rc script style, sample handling, and the brutefir dependency; having the Phase 0 fixes merged upstream is the strongest argument the stack works on stock FreeBSD. Ongoing: bump the port per release, watch fallout from musicpd/upmpdcli/virtual_oss updates.

10.6 Recommended order of value

1. **Phase 0 upstreaming** — benefits every FreeBSD USB-audio user, already in flight, and a hard prerequisite anyway.
2. **Phase 1.2 (flat filters)** — small, immediate, makes the repo usable by others today even without a port.
3. **Rest of Phase 1** — worthwhile for the repo's own health regardless.
4. **Phases 2-3** — only once the BruteFIR question is settled and there is evidence of an audience; a port is a maintenance promise.

11 Appendix C — Bit-perfect test assets and cross-OS comparison

The Tools chapter (*Bit-perfect verification*) covers the single-host proof. This appendix documents the test assets and the cross-OS procedure that proves the Linux and FreeBSD boxes send the DAC the *very same bytes*.

11.1 Test assets (tests/)

Two deterministic, near-silent (~ -90 dBFS) signals; every sample is uniquely determined, so any truncation, dither, volume change, resampling or channel swap shows up immediately.

- **Short asset (committed)** — `bitperfect-test-44100-s32-stereo.wav`: S32_LE, 2 ch, 44100 Hz, 100000 frames (~2.27 s). Per-sample counter in the low 16 bits, $L = i \& 0xFFFF$, $R = (i * 40503) \& 0xFFFF$. Its PCM payload is byte-identical to the reference `.raw` (the WAV is the same bytes plus a 44-byte header — MPD cannot play headerless raw).
- **Cross-OS asset (generated, not committed)** — `bitperfect-test-44100-s32-stereo-30s.wav`: S32_LE, 2 ch, 44100 Hz, 1323000 frames (30 s), 10.6 MB. Here *every* (L, R) pair is unique over the whole file ($R = (i * 40503 + (i >> 16)) \& 0xFFFF$ folds the block index in, breaking the 65536-frame period), so capture alignment is unambiguous at any length. It is too large to commit; regenerate it byte-identically on any OS:

```
python3 tests/gen-bitperfect-wav.py \
    tests/bitperfect-test-44100-s32-stereo-30s.wav
# sha256 88d365eeacbb1fa830bb1a2726b0f29bb545885824351080e0c5b4cbc9602348
```

11.2 Other rates and sample widths

The generator takes `--rate`, `--bits` (16/24/32) and `--frames` (= seconds x rate), so any format the DAC advertises can be tested. Both tap scripts read rate and width from the WAV header — feeding them a different file is the whole configuration:

```
python3 tests/gen-bitperfect-wav.py --rate 192000 --bits 24 --frames 5760000 \
    tests/bitperfect-test-192000-s24-stereo-30s.wav
./scripts/bitperfect-tap-linux.sh tests/bitperfect-test-192000-s24-stereo-30s.wav
```

The sample *values* are the same counter at every width (they never exceed 0xFFFF), so only the container changes — which means a 16/24-bit asset additionally exercises the **lossless promotion to the 32-bit USB wire container** (<<8 for 24-bit, <<16 for 16-bit) that any bit-perfect player must perform for a DAC accepting only 32-bit containers. A 16/24-bit input therefore does not change the playback path at all: prep promotes first and the player always emits S32_LE.

Two consequences: **each format has its own sha256** (a 24-bit file stores 3 bytes per sample, so it is a different file), and a cross-OS comparison must use the **same width on both machines** — differently shifted wire values never byte-match. The report's `ref` bytes hash is that of the promoted stream, so it too differs per width at the same rate.

Rate	Bits	Frames	Size	sha256 (first 16)
44100	32	1323000 (30 s)	10584044	88d365eeaccb1fa8
44100	24	1323000 (30 s)	7938044	e2702c119606cdf8
192000	24	5760000 (30 s)	34560044	58dd87f3560334fb
192000	24	1920000 (10 s)	11520044	01317af6523ec67f
96000	24	960000 (10 s)	5760044	b572faabdee3b623

Any other combination is equally valid: the generator prints the sha256 of whatever it writes — generate once, note the hash, match it on the other machine (full hashes in `tests/README.md`). `.gitignore` excludes the generated assets by name (the 44100/32-bit and 44100/24-bit 30 s files and the 192000/24-bit 10 s one), so add any further WAV you keep, or drop it under `bp-results/` (ignored except for `*.txt`).

11.3 Verification status

Both taps are executed and passing — the FreeBSD side is no longer a written-but-unrun script:

Host	Runs	Result
Linux (DacMagic 100, kernel 7.1.5-arch1)	44100/32-bit x 30 s, 44100/24-bit x 30 s, 192000/24-bit x 10 s	all BIT-PERFECT , 0 truncated events, 0 usbmon drops
FreeBSD 15.1-RELEASE (same DAC, <code>usb0 devaddr 2</code>)	same three	all BIT-PERFECT (exit 0); the 192 kHz run confirms the DAC clock followed (<code>dev.pcm.0.feedback_rate = 191994</code>) and costs ~24 s wall clock

`bitperfect-compare.py` reports **MATCH** across the two hosts for the 44100/32-bit asset; the 24-bit pairs have no committed Linux counterpart yet, so those stand as local per-host proofs. The comparator itself has been exercised on every path (`wav/wav`, `wav/txt`, `txt/txt`, refusal of `raw/txt`) plus a deliberately bit-flipped payload, which it reports as **MISMATCH** at the exact offset.

The FreeBSD tap's first run exposed a real defect — a capture truncated by ~17 ms because `usbdump` discards its unflushed buffer on exit. The fix is a 500 ms **silence pad**: the player plays `ref.raw` plus half a second of zeros, while the verdict still compares the unpadded reference, so the loss lands in silence nothing depends on. The pad is removed by *arithmetic* (take `len(ref)` bytes from the alignment point), never by silence detection — the test signal is itself near-silent, so a zero-seeking trim would eat real payload.

11.4 Cross-OS byte comparison

Each OS taps its own USB isochronous OUT endpoint while playing the (locally regenerated) common WAV, then the reports are compared. Only the tiny `bp-results/*.txt` reports are committed — they carry the tap payload's length and sha256, which proves byte-identity without moving the 10 MB streams through git.

```
# Linux box:
./scripts/bitperfect-tap-linux.sh \
  tests/bitperfect-test-44100-s32-stereo-30s.wav
git add bp-results/*-linux.txt && git commit && git push

# FreeBSD box (free the DAC first: ./drc.sh off):
./scripts/bitperfect-tap-freebsd.sh \
  tests/bitperfect-test-44100-s32-stereo-30s.wav
git add bp-results/*-freebsd.txt && git commit && git push

# then on either box:
git pull
./scripts/bitperfect-compare.py \
  bp-results/bitperfect-test-44100-s32-stereo-30s-linux.txt \
  bp-results/bitperfect-test-44100-s32-stereo-30s-freebsd.txt
```

Identical length and sha256 on both reports proves the two operating systems deliver bit-identical audio to the DAC. Step-by-step commands and the mismatch-forensics path are in `scripts/README.md` and `doc/BIT-PERFECT-VERIFICATION.md` (*Cross-OS comparison*).

A single run already proves the local path. Each tap compares its capture against the reference derived from the input file *on that machine* and exits 0 on **BIT-PERFECT** — a complete file -> USB proof by itself. `bitperfect-compare.py` is a separate, optional step answering the further question of whether two hosts agree.

What the report records. Each `PREFIX.txt` names and hashes every stage (input file, ref bytes, wire raw, tap wav, verdict), so a run is auditable from the ~600-byte report alone. Only three of those are reproducible: input file, ref bytes and tap wav. **wire raw is not** — it is the untrimmed capture, so its length varies between otherwise identical runs (a few packets more or fewer recorded before the tap stops), and Linux and FreeBSD captures of the same input legitimately differ (10584816 vs 10772776 bytes at 44100/32-bit). It is provenance only; the field the comparator uses is tap wav. Note also that `PREFIX.wav` equals the input WAV only for a **32-bit** input: for 16/24-bit it carries the promoted 32-bit container, so it is longer and differently valued — the invariant that always holds is the tap wav payload hash, not the file hash.

What surrounds the audio. The capture is always longer than the reference, and everything outside is measurably all-zero: a head of stream-priming zeros — **exactly 16 ms at both 44100 and 192000 Hz**, a fixed-duration buffer prime rather than a timing accident — and a tail of the 500 ms pad plus ~19 ms the kernel keeps transmitting after the writer closes. On a **BIT-PERFECT** verdict both capture boundaries necessarily fell outside the audio; when they fall inside, the tool names it (HEAD LOST at the start, INCOMPLETE at the end) rather than hiding it. One qualification: “inaudible” describes the sample *values*. Opening or closing an isochronous stream, and any rate change around it, can still produce an audible artifact from the DAC’s analogue side (mute relay, PLL relock — see OKT0-DAC8-FreeBSD-44k1-flicker.md); that comes from stream start/stop, not from the zeros.

Start with the canonical 44100 Hz asset on FreeBSD. The FreeBSD tap decodes `usbdump -vv text`, so parsing cost scales with the capture: 30 s at 44100 Hz is ~10.5 MB of payload arriving as tens of MB of hex-dump text (fine), while 30 s at 192 kHz is ~46 MB of payload as several hundred MB of text — slow, and it stresses the pcap capture too. Prove the path at 44100 first, then shorten the high-rate run (`--frames 1920000` = 10 s at 192 kHz). The Linux side reads `usbmon`’s binary interface and has no such limit.

12 Appendix D — Source document index

This manual is generated from the repository's source files — the Markdown docs below plus the CMake build for the Installation chapter. For the full detail behind each section:

Topic	Source document
Chain overview, install, drc.sh, hotplug	README.md
Install / build (CMake superproject, host values)	CMakeLists.txt, host.cmake.sample
Build modules: DRC engine + site data	cmake/core-drc.cmake
Build modules: DAC-hotplug + brutefir services	cmake/hotplug.cmake
Build modules: MPD + upmpdcli renderers	cmake/renderers.cmake
Build modules: runtime dependency audit	cmake/dependencies.cmake
Build modules: per-user setup (make user-install)	cmake/user-install.sh.in
Web-UI subproject builds	omdrc-ctrl/CMakeLists.txt, video/webremote/CMakeLists.txt
Filter/config layout, drc.sh modes, agent rules	FILTERS_AND_DRC.md
Filter provenance, hashes, verification, the design scripts	doc/FILTER_PROVENANCE_AND_RESPONSE.md
Site-data split (OMDRC_SITE_DATA_DIRS / OMDRC_SITE_ROOT)	scripts/README.md, host.cmake.sample, cmake/core-drc.cmake
Helper scripts	scripts/README.md, README.md
Web control panel	omdrc-ctrl/README.md
CD / S-PDIF input bridge	cdin/README.md
ESI U24 XL configuration and traps	cdin/ESI-U24XL.md
Stable sound-device names and lifecycle	etc/rc.d/omdrc_audio, etc/devd/omdrc-audio.conf
Spectrum analyzer	omdrc-ctrl/SPECTRUM_ANALYZER.md
Video playback + Blu-ray	video/README.md
A/V sync delay derivation	video/AV-SYNC-DELAY.md
Web remote (install/API)	video/webremote/README.md
Web remote (design)	video/webremote/ARCHITECTURE.md
Glitch detection	doc/GLITCH-DETECTION.md
Bit-perfect verification	doc/BIT-PERFECT-VERIFICATION.md
Test signal	tests/README.md
FreeBSD port plan	doc/FREEBSD-PORT-PLAN.md
uaudio patches (index + install)	freebsd-uaudio-patch/README.md
44.1 kHz flicker analysis	freebsd-uaudio-patch/FreeBSD-uaudio-shared-clock-bug.md

Topic	Source document
Shared-clock fix design	freebsd-uaudio-patch/uaudio-shared-clock-fix.md
Clock-before-alt analysis	freebsd-uaudio-patch/uaudio-clock-before-alt.md
Feedback-follow audit	freebsd-uaudio-patch/uaudio-feedback-follow.md
virtual_oss patches (index)	freebsd-virtual-oss-patch/README.md
SETTRIGGER livelock root cause	freebsd-virtual-oss-patch/ROOTCAUSE-settrigger-sync-engine-deadlock.md
cuse reflink root cause	freebsd-virtual-oss-patch/ROOTCAUSE-cuse_client_open-reflink.md
cuse teardown bug report	VIRTUAL_OSS_CUSE_DEADLOCK.md
Cold-open silence audit	OKT0-DAC8-silent-first-open.md
44.1 kHz flicker observations	OKT0-DAC8-FreeBSD-44k1-flicker.md
MPD/curl CPU spin	MPD-CURL-CPU-SPIN-FreeBSD.md
Kodi OSS sink patch	kodi-virtual-oss-patch/README.md
VBA vs all-pass comparison	doc/xtras/FVBA.vs.ALLPASS.md

12.1 Keeping this manual up to date

This manual is a **synthesis** of the source documents above, not a transclusion: rebuilding the PDF does *not* pull in changes to the source .md files automatically. When a source document changes:

1. Update the corresponding section of the master document, doc/pdf/open-media-drc-manual.md (the table above is the section-to-source mapping).
2. If the chain topology changed, adjust the graphviz sources in doc/pdf/diagrams/*.dot.
3. Re-run doc/pdf/build-pdf.sh (requires pandoc, pdflatex, graphviz) to regenerate doc/open-media-drc-manual.pdf.

Editing constraints for the master document (pdflatex): keep it ASCII — no box-drawing characters or Unicode arrows/symbols; diagrams are added as `![caption](build/<name>.svg){width=NN%}`. Full details in doc/pdf/README.md.